

3.1

Introduction—The Basic Concept

The 224X is a sophisticated signal processor, in both its capabilities and control functions. It is much more versatile than other reverberation devices, and the LARC increases its ease of use even further by offering an interactive, prompting interface. A system of banks, programs and variations, control pages, and registers makes this versatility immediately available, even to inexperienced users. You will find the LARC a powerful and exciting addition to the 224X, and if you are a new 224X user, you will find the LARC an informative, self-prompting interface into the world of digital reverberation. Whether or not you are familiar with the 224X, please read this section to understand how to operate the LARC and 224X.

The 224X digitizes incoming signals and processes the digitized signals; it then reconverts the processed signals into analog for output. Because the modifications to the sound are totally under the control of the 224X's high-speed processor, which is directed by the LARC and its software, the range of sound is extremely wide. The 224X has the ability to emulate the response of many different kinds of actual spaces, mimic other artificial reverberation devices, and create complex effects with minimal controls. In addition, the Split programs allow the 224X to serve as two separate reverb devices.

3.2

Banks, Programs, Pages, and Variations

3.2.1 How To Load Programs

The 224X with LARC stores its reverberation programs in banks. A bank is a convenient grouping of similar, but unique, programs comprising sets of instructions that direct the 224X's high-speed processor to emulate acoustical spaces, special effects, or both. Each program defines a set of variable parameters that uniquely characterize the program.

For convenience, all banks and programs have names, and all programs have at least one preprogrammed variation—groupings of preset parameters. Changing parameters is accomplished through control pages, which are arrangements of variable parameters whose values can be changed by moving the LARC's sliders—most programs have four control pages, and each page holds as many as six variable parameters (corresponding to the LARC's six sliders).

By pressing combinations of the LARC's upper rows of labeled pushbuttons, you can access the banks, programs, variations, and control pages. (You are automatically placed in page 1 of variation 1 when you access a program.) When in a variation, you can alter parameter settings using the LARC's sliders, tailoring each variation to suit your taste and applica-

tion. If you wish to save your settings for future use, you can store them in one of 36 memory registers. If power is interrupted or the unit is reset, the 224X brings up the last active bank, program, and variation, including all the settings then in effect.

In general, any variation within a given program can be generated from any other variation simply by changing the operating parameters with the control sliders. One of the best ways to get acquainted with the 224X is to store a variation in a register, then call another variation of the same program and modify it until it sounds exactly like the first one. As long as the two variations come from the same program, this procedure can always be performed.

The LARC allows the greatest flexibility of control over the 224X. It can be used to scroll through (without loading) possible banks and programs by stepping through menus, or it can serve as a command console to load a desired program, variation, or register instantly. The LARC's upper display window guides you through these operations.

Operation

To load programs:

■
from the current bank: press PROG, N (where N is one of the numeric-select keys from 1 to 0)

■
from a different or another bank: press BANK, N, PROG, N.

Note: *The BANK and PROG keys can scroll through their respective contents (displaying titles only) without loading programs; for example:*

■
BANK, BANK, BANK scrolls through three banks

■
BANK, 3, PROG, PROG scrolls to the second program in bank 3.

Note: *Once PROG has been pressed, the numeric-select keys can be used to load any of the programs in a given bank.*

Remember that a program will run only when pressed in conjunction with a numeric-select key; for example:

■
BANK, 2, PROG displays the name of bank 2 program 1; to run this program, press the numeric-select key 1.

When a program is active, a banner telling you what bank, program, and variation are running appears in the bottom right of the LARC's upper display. The banner looks like this:

B5 P4 V1

At any time, you can display this banner by pressing VAR once, or one of the keys under the sliders; after pressing a slider key, the slider's value is displayed and after a few seconds, the banner appears.

To access variations within a program:

■
press VAR, VAR, ... VAR, which sequentially steps through all available variations (unlike the BANK and PROG keys, VAR activates the variations in sequence)

or

■
press VAR, N, which selects variation N.

Note: *Once VAR has been pressed, the numeric-select keys can be used to activate any of the variations. Pressing N, N, ... N changes to other variations.*

When you become familiar with the system of banks, programs, and variations, shuttling between them is easy; for example:

To load bank 3 program 2 variation 4:

■
press BANK, 3 (or successively press BANK until BANK 3 appears in the LARC's upper window)

■
next press PROG, 2 then VAR, 4 (or successively press VAR four times).

3.2.2 How To Modify Programs

After loading a program, you can adjust and modify its variable parameters by moving the LARC's sliders. Sliders are grouped in convenient control pages; to access these control pages:

■
press PAGE, N or PAGE, PAGE, ... PAGE, which successively steps through each control page.

When a new program, variation, or control page is selected, the sliders are deactivated (the display does not change) until they are moved through their preset values. When

changes have been made on a page, and a new page is activated, those previous changes remain intact and disappear only when another variation or program is loaded. Because all variations have preset values (remember that you are automatically placed in page 1 of variation 1 when you load a program), you must first move a slider through its preset value to activate it. (You can activate all sliders on a given page without passing their preset values by pressing 2nd F, PAGE.) Anytime a slider is moved, it displays its values in the LARC's upper window while it is

being moved. Sliders can be adjusted in real time—tuning a desired reverberation or special effect.

Note that abbreviated codes appear in the display window above active sliders. These codes indicate the parameters that the sliders are controlling. To receive a more descriptive title for each parameter, press the keys directly

below each slider. This procedure displays the full parameter title and its current value in the LARC's upper window. Also, this display appears whenever a slider is moved.

Section 3.3 describes in greater detail how the variable parameters alter the programs.

3.2.3 How To Store and Recall Modified Programs

Once satisfied with your modified settings, you can store them in one of the 224X's 36 memory registers. The registers are organized into banks that can be loaded on tape via the LARC for storage.

Register Storage. The 36 registers can be divided among 10 possible banks; each bank can hold as many as 10 registers.

To access register banks, press REG. You can now scroll through available register banks by pressing BANK, BANK, ... BANK or press BANK, N to access a particular bank.

To exit from the register menu and return to the program menu,

■
press VAR.

Once you have reached the desired bank, you can scroll through the registers in the bank by pressing REG, REG, ... REG. (Pressing REG, N loads a register that has been previously stored.)

To store modified parameter settings in a register, once you have scrolled to the appropriate bank and register

■
press STO and while you hold it down the display will read:

STORE BX RY

where X and Y are the numbers of the last scrolled bank and register, respectively.

The register is stored when you press REG while still holding down STO, and the LARC displays the derivative program on the top line. The bank and register numbers of the

stored settings are displayed on the bottom line, and the derivative bank and program appear in brackets. For example:

CONCERT HALL

B3 R5 [B1 P1]

This example signifies that a derivative of the Concert Hall program (bank 1 program 1) is now stored in register bank 3 register 5. Remember that when you store settings in a register, the previous contents of the register are erased. Note that until a stored register is labeled by a user, it is automatically called -- --; these hyphens indicate spaces available for labeling, the procedure for which is discussed in Sec. 3.2.4.

To from exit a running register and return to programs

■
press PROG, N.

To clear a register

■
press 2nd F, STO and while holding down STO, press REG.

When a register has been cleared, it is relabeled EMPTY.

Note that when all 36 registers are used, attempting to store another register results in the message:

NO ROOM LEFT

IN REGISTERS

Register Recall. Once registers have been stored, they can be recalled and activated by scrolling to the appropriate register bank and pressing REG, N. Once in the register mode, various registers in the same bank can be accessed by pressing N, N, ... N.

3.2.4 How To Label Banks and Registers

To label a register bank

■ scroll to the bank you wish to label

■ press 2nd F, BANK.

The display will read:

ENTER LABEL

FOR BANK X

where X is the number of the bank being labeled.

All six sliders can now be used to label bank X. (42 characters including a space are available on each slider.) Once the desired label is obtained, press BANK to store it. A bank retains its user label until changed using this procedure, even if all registers in the bank are empty.

To label a register

■ scroll to the register you wish to label

■ press 2nd F, REG.

The display will read:

ENTER LABEL

FOR REG X

where X is the number of the register being labeled.

The first five sliders can be used to label the register. Once the desired label is obtained, press REG to store it. A register is automatically relabeled EMPTY when it is cleared and cannot be relabeled until something is again stored in it.

3.2.5 Tape Storage and Recall

Equipment Required. To store register banks on audio tape, use the following equipment:

■ high-quality audio tape recorder (cassette, microcassette, or reel to reel) with earphone, external speaker, or line-level output jack and an auxiliary or line-level input jack—a microphone input is suitable, providing an appropriate attenuator is placed between the input plug and jack

■ Lexicon cassette interface cable (Lexicon no. 680-03690) with a male 5-pin 180° DIN connector terminating in two miniphone plugs (cable also available from Radio Shack cat. no. 26-1207), supplied with the LARC

■ high-quality audio tape (such as TDK or Maxell); if cassettes are used, avoid tapes longer than 60 minutes per side.

Note: *The tape recorder/playback unit must provide good performance and accurate head-to-tape contact, tape alignment, and speed to ensure interchangeability between different tape decks. During recording, use the Automatic Level Control (ALC or AUTO) or set the VU meter to 0 if using a manual control. (For cueing and level settings, the LARC inserts a 15-second 4800-Hz tone leader before and a 3-second trailer tone after each entire data transmittal. The 15-second leader is made up of 3 seconds of pure tone and 12 seconds of pure tone with intermittent clicks.)*

If the playback unit is equipped with a tone control, adjust it for reduced high-frequency response to prevent ringing in the playback signal. The nominal tape record output from the LARC is +4 dBm (equivalent voltage into 600 ohms), and the tape playback level to the LARC can be between 0 and +8 dBm. Adjust the playback output level to +4 dBm (or adjust for reliable operation if monitoring is not convenient). With good tape and players, playback level should not be critical.

Connections. Connect the male 5-pin DIN plug on the Lexicon cassette interface cable into the LARC's rear panel. Connect the black miniphone plug into the tape recorder's earphone, external speaker, or line-level output jack, and connect the gray miniphone plug into the auxiliary or line-level input jack (a microphone input can be used with suitable attenuation).

Note: Check all connections before storing data.

Tape Storage. Once the appropriate connections have been made, ready the tape deck by (1) loading and cueing the tape, (2) switching (if possible) the input level control to AUTO, and (3) adjusting the tone or treble control to low or minimum. (If the input level cannot be set to AUTO, set the VU meter to 0 during the first 15-second leader tone just before data are transferred to tape.)

To store a register bank or banks:

■
press TAPE three times to select the prompt:

STORE BANKS

ON TAPE

■
select the bank (or banks) you wish to store by pressing the corresponding numeric-select key or keys in the order you wish to store them—the numbers selected appear in the bottom left of the display. The register banks are recorded onto the tape in the order chosen. Be sure to enter the same banks in the same order that they were stored in.

Start the tape deck in record mode.

■
press TAPE to commence output of data to tape.

(If you wait longer than 45 seconds, the LARC returns to the last active program and cancels the storage process. To cancel the process at any time, press VAR.)

In the lower right of the display, a status indicator flashes during the storage process; note that the LARC inserts a 15-second leader at the beginning and a 3-second leader at the end of the complete data transfer.

When the transfer is complete, the display will read:

TAPE STORE

COMPLETED

After a tape transfer is complete, always verify that the data stored are accurate.

Tape Verification. Verify that all data stored are complete with the following procedure:

■
press TAPE.

The display will read:

VERIFY BANKS

AGAINST TAPE

■
press the appropriate numeric-select key or keys for the banks you wish to verify (in the same order in which they were stored).

Cue the tape to the beginning of the segment you wish to verify.

■
press TAPE.

Start the tape in playback mode. Once the verification is complete, the display will read:

VERIFY GOOD

000SE 000HE

or

VERIFY BAD

002SE 001HE

The bottom line indicates soft (correctable) and hard (uncorrectable) errors that have occurred during the transfer. Soft errors (SE) do not affect the data and are displayed for diagnostics only. Hard errors (HE) indicate problems with the transfer that require repeating the process. (The example shows 2 soft and 1 hard error.) In these cases, try the verification procedure again. If it is still bad, re-store the register banks and repeat the verification procedure.

Note: *Continued occurrence of soft errors could indicate a deficiency in the system—possibly an inferior tape deck or tape, improper record or playback levels, or excessive high-frequency response.*

When the the storage is verified, press VAR.

Tape Recall. To recall register banks from tape storage, use the following procedure:

■ press TAPE twice.

The display will read:

RECALL BANKS

FROM TAPES

■ press the numeric-select keys of the banks you wish to fill (the banks stored on tape are recalled in the order in which they were stored—their previous numbers are not carried over).

Cue the tape to the beginning leader of the segment you wish to recall.

■ press TAPE.

Start the tape recorder in playback mode. When the recall is complete, the display will read:

RECALL GOOD

000SE 000HE

or something like

RECALL BAD

004SE 001HE

As with tape verification, the bottom line indicates errors that may have occurred during the process. Soft errors do not affect the recall, but hard errors indicate that the information stored in the LARC's register banks does not coincide with the information stored on the tape. Check the playback volume and tone control on the tape deck, and repeat the recall.

The LARC recalls for storage as many registers as space allows. If the total is more than 36 registers, the LARC fills up the last available register until the limit of 36 registers is reached.

Note: *To ensure safekeeping, store important data twice on tape and, if possible, record a backup tape.*

3.3

How The Controls Affect The Sound

This section discusses the functions of the main reverberation parameters that are stored in the 224X. Those parameters that apply to the Effects programs are discussed in detail under their specific programs in Sec. 4.

Table 3.1 briefly lists and describes all variable parameters and toggles including those applicable to the Effects programs.

Table 3.1. Control Parameters.

Parameter	Description
Reverberation Parameters	
Attack	Sets the explosiveness or sharpness of the initial response to an input. (On Plate and Chamber programs only.)
Chorus	Modulates the delays in reverberation programs or the Chorus & Echo program. Can introduce more fullness to sound, create the illusion of added voices, or produce flanging and pitch-twisting effects.
Crossover	Controls the frequency at which a transition from low-frequency reverb time to mid-frequency reverb time takes place.
Decay Optimization	A parameter toggle that activates and deactivates a subroutine that automatically optimizes decay parameters for various source levels and material.
Definition	Controls the echo density in the latter part of the sonic decay.
Depth	Controls the illusion of position within an acoustic environment. (On Hall and Room programs only.)
Diffusion	Controls the degree to which echo density increases over time.
Dynamic Decay	A parameter toggle that activates and deactivates the LF and Mid Stop Decay controls.
Fine Predelay	Finely adjusts the predelay values independently in each of two stereo channels.
Gate	Sets a time delay in switching from running decay to stopped decay (when using the dynamic decay feature). Used for gated reverb effects.
HF Bandwidth	Sets the corner frequency of a 6-dB/octave low-pass filter that affects all sound from the 224X.
LF Decay	Sets the reverberation time of low-frequency signals.
LF Stop Decay	Sets the decay time of low-frequency signals in the absence of input.
Mid Decay	Sets the reverberation time of mid-frequency signals.
Mid Stop Decay	Sets the decay time of mid-frequency signals in the absence of input.
Mode Enhancement	A parameter toggle that activates and deactivates the Chorus control.
Predelay	Sets the time from an input to the onset of reverberation.

Table 3.1. Control Parameters cont'd.

Parameter	Description
Reverberation Parameters (Continued)	
Preecho Delay	Sets the delay time of the early echoes.
Preecho Level	Controls the level of the early echoes.
Size	Sets the apparent size of the acoustical space produced by the 224XL (essentially the longest dimension in meters).
Slope	Controls the decay characteristics of the Inverse Room program.
Treble Decay	Sets the frequency above which sounds decay at a progressively faster rate.
Effects Parameters	
Band Delay	Sets the delay time for each frequency band in the Multiband Delay program.
Band Level	Controls the relative level of each frequency band in the Multiband Delay program.
Crossfeed	Controls the level and balance of a feedback mix routed back into the Left and Right input channels in the Resonant Chords program.
Feedback	Controls the level and polarity of signals recirculated back into the delay line in the Chorus & Echo program (and two delays in the Multiband Delay program).
HF Cutoff	Sets the high-frequency cutoff of a low-pass filter (6 dB/octave).
LF Cutoff	Sets the low-frequency cutoff of a high-pass filter (6 dB/octave).
Note Level	Sets the level of the notes in the Resonant Chords program.
Note Pitch	Tunes each note in the Resonant Chords program to a desired pitch.
Note Predelay	Sets the time from an input to the onset of a note in the Resonant Chords program.
Pan	Places the sound of the separate channels along a left-to-right continuum within a stereo field. (Resonant Chords, Multiband Delay, and Chorus programs.)
Resonance	Regulates the amount and polarity of feedback to the individual delay lines in the Resonant Chords program.
Voice Delay	Controls the delay time of each of the "voices" or delay taps in the Chorus and Echo program.
Voice Level	Sets the level of each of the "voices" or delay taps in the Chorus and Echo program.

3.3.1 Variable Reverberation Parameters

Decay Time vs. Frequency: LF and Mid Decay, Treble Decay, and Crossover.

Reverberation is usually greatest right after an input, and then it decays approximately linearly down to zero. The time required for reverberation to decay 60 dB from its peak value is referred to as RT60. The overall reverberation time and the ratio of reverb to original sound (set by the mixing console) determines whether a sound is wet (reverberant) or dry (acoustically dead). Decay times for normal reverberation range from a fraction of a second to several seconds. In addition to duplicating a wide variety of natural reverberation phenomena, the 224X can extend decay times far beyond the realm of natural phenomena to produce a range of interesting effects.

Under natural acoustic conditions, different portions of the audio spectrum decay at varying rates, so that the tone of the reverberation actually changes over time. The 224X provides extensive control over the tone of reverberation, allowing decay time to be set in relation to frequency. In the reverberation programs, the LF and Mid Decay sliders control decay time (expressed in RT 60) of the low and middle portions of the frequency spectrum, respectively. The Crossover slider adjusts the point in the frequency spectrum in which the LF Decay slider relinquishes control and the Mid Decay slider assumes control. The Treble Decay slider adjusts the point in the frequency spectrum above which the frequencies fade away at a much faster rate than the lower frequencies. The Treble Decay slider, therefore, sets the frequency above which decay time becomes progressively shorter.

Choice of decay time depends on material and desired ambiances or effects. Shorter decay times maintain articulation and impart an "up close" ambience to the sound. As decay times are lengthened, the sound begins to take on more of a texture of "lushness." Decay times affect different sounds in different ways. Classical or symphonic music is typically produced in environments having decay times from 1.8 to 2.5 seconds, whereas organ music or religious music produced in a church have typical decay times of several seconds. Although this reverberation sounds normal for organ or religious music, if a vocal in popular

music were to have the same decay characteristics, it would sound exceptionally rich.

As an example of how these controls are used together to create different ambiances, in the Concert Hall program, a really high-quality concert hall can be created by setting LF Decay between 2.5 and 3.0 seconds, Mid Decay between 1.8 and 2.5 seconds, and Crossover between 600 and 800 Hz (as in variation 1). A typically poor concert hall sound can be obtained by setting LF Decay to 1 second, Mid Decay between 1.5 and 3.0 seconds, and Crossover to 1000 Hz. The lack of low frequencies in relation to the mid frequencies in the decay of the poor concert hall makes the sound seem thin, without body or warmth—reminiscent of modern concert halls whose reflecting surfaces are light-panelled, thinly constructed walls.

To duplicate the natural phenomenon of air absorption of high frequencies, the Treble Decay control should always be set between 6 and 10 kHz. For the previous examples, 6 kHz should provide the most natural setting.

For special effects, and for rock and other types of popular music, experimentation with different settings of these controls in a variety of combinations can lead to finding a special ambience, effect, or sonic shading for a specific application. In multitrack recording, it is not uncommon to create a different ambience for each track or instrument/vocal section to lend a distinct color to each for a rich, varied sonic impression. The Split programs in the 224X are particularly convenient for this application because they allow two totally distinct sounds to be used on different tracks simultaneously without rerecording.

Dynamic Decay: LF and Mid Stop Decay.

In addition to the regular decay characteristics set up by the LF and Mid Decay, Treble Decay, and Crossover controls, decay characteristics that apply only to the end (tail) of a sound can be set up using the LF and Mid Stop Decay controls. These controls are analogous to the LF and Mid Decay sliders, except that they are activated by the Dynamic Decay toggle and they affect reverberation only after the input sound has stopped.

When the Dynamic Decay toggle is turned on, the LF and Mid Stop Decay sliders can set the amount of reverberation in the respective bands DURING PAUSES BETWEEN SIGNALS INPUT INTO THE 224X. The regular LF and Mid Decay sliders set the decay when input is present. If running decay is short and stopped decay is long, the sound will have crisp, clean, well-articulated and defined attacks with long, flowing echo tails.

Appropriate source material is highly critical in achieving this effect. Lead vocals or instrumental solos with a somewhat continuous flow and distinct breaks between phrases are ideal. These materials enable each phrase to be cleanly articulated, while providing echo trails at the ends of the phrases for a defined but "fat," "wet" sound. A full ensemble mix will usually swamp out the dynamic effect because phrase endings in individual lines are masked by the continuing sound of other lines. Choppy input material may also produce inappropriate results because the brief pauses within a particular passage can be mistakenly interpreted as phrase endings.

If stopped reverb decay is less than running reverb decay, the effect is reversed. Phrases in backup material can be placed in pools of reverberation, yet be cleared out of the way of new phrases in lead material. The results can be very startling! With percussion, this feature can be used to "gate" the reverb, so that a drum hit could result in an enormously fat sound of short duration to maintain the tight, percussive quality of the drum.

Predelay. Predelay is the amount of time that elapses from an input to the onset of actual reverberation. Under natural conditions, the amount of predelay is a function of the size of the acoustic space and the relative positions of the sound source and listener(s). The Predelay slider on the 224X duplicates this phenomenon and is used to create a sense of distance and volume within an acoustic space. A long predelay places the reverberant field BEHIND rather than on top of the input. Many of the 224X programs have a lower limit on the amount of predelay determined by the size of the acoustic space being represented. Some programs have Fine Predelay controls for the left and right channels that adjust the relative time of the first reflections; these can be used to fine tune the stereo image of the reverberation image.

A sense of continuity between source and reverb is maintained up to around 40 milliseconds of predelay, after which sound begins to break up into distinct "slap" echoes; however, large values of predelay effectively give the impression of large size if early reflections (pre-echoes) are used to fill in the spaces between the input and the delayed reverberation.

Depth. The Depth slider (on the Halls and Rooms programs) controls the illusion of position within the reverberant field. Visualize a concert hall. If a listener is seated in the first row in the concert hall, the reverberant sound is primarily reflections from the front wall (stage back), with few reflections from the side walls and even fewer reflections from the rear.

As the listener moves farther and farther back, the proportions of the different reflections change. The Depth control helps create the illusion of different positions within an acoustical environment. When the Depth slider is fully lowered, the reverberation output is composed almost entirely of early reflections, representing a position close to the stage. As the control is raised, greater proportions of side and rear wall reflections are heard, representing positions farther to the rear of the concert hall, until the reverberation output is composed almost entirely of rear wall reflections.

Attack. This control (on the Chamber and Plate programs) sets the level of the initial sonic pulse (attack). Attack affects the level of sound within the first 50 milliseconds only; high settings cause an explosive sound, and low settings cause the sound to build up more slowly with time.

Chorus. Advancing the Chorus control makes reverberation sound less metallic by randomizing delay times in the reverberation programs. This control is normally active, but can be turned off with the Mode Enhancement toggle. Raising or lowering the Chorus control varies the rate of modulation. Because the control causes pitch variation, sources with very little pitch wobble, such as guitar or piano, should have chorus values at or below 50; vocals can use 50 to 60; and spoken voice up to 70. A good practice is to raise the slider until pitch wobble becomes noticeable, then lower it slightly. Although the effect of the Chorus control is quite different for the reverberation

programs and the Chorus and Echo program, the principle is the same (for more detail, see Sec. 4.4.1).

HF Bandwidth. This control sets the frequency above which a low-pass filter attenuates all tones by 6 dB per octave. This attenuation occurs at the input of the reverberation algorithm and affects the preechoes as well as the reverberant sound. This control does not affect the rate at which high frequencies decay, which is determined by the Treble Decay slider.

Diffusion. To understand Diffusion, consider the output of the 224X in response to the input of a single click. After a delay time set by the Predelay slider, the single-pulse input produces multiple pulses in the output, decreasing in amplitude and increasing in density with time. Thus, when a click is reverberated, one first hears a few relatively discrete reflections that soon blend into a smooth swish. Diffusion refers to the degree to which the density of echoes increases. This parameter is affected by program selection and is controlled by the Diffusion slider.

The Diffusion slider sets the amount of initial buildup in density. High settings of Diffusion result in high initial buildup, and low settings cause low initial buildup. After the initial period controlled by the Diffusion slider, density continues to rise at a rate determined by the program. For example, in the Concert Hall program, the density starts building at a slow rate, but then rapidly increases. In the Constant-Density Plate A program, the density remains constant after the initial buildup period, and in the Rich Chamber program, there is a slow, continual buildup after the initial buildup set by the Diffusion slider.

Much of the difference between programs is because of the different rates at which density increases. The Hall, Plate, and Room programs use a rapid increase in density with time, so even when Diffusion is set low, after a time (approximately 300 milliseconds for the Hall and Plate programs, 150 milliseconds for the Room program, and 70 milliseconds for the Small Room program), the density is great enough that individual echoes cannot be distinguished. In contrast, the Constant-Density Plate programs maintain a constant density after the initial buildup determined by the Dif-

fusion control. If Diffusion is set too low in this program the reverberation has a noticeably grainy quality that does not smooth out as the sound decays. Density in the Rich Chamber program increases with time, but at a much lower rate than that of the Hall or Plate programs, and this low rate of increase gives this program less "color" as the sound decays. Low diffusion is good for vocals, and high diffusion is good for percussion.

Definition. This control affects the rate of increase in density in the latter part of the sonic decay. In the lowest position of the slider, the rate is that determined by the program; raising the slider decreases the rate. Raising Definition causes the sound to become choppy as the decrease in density of the echoes creates increasingly distinct repetitive echo trails rather than a smooth, continuous fusion of sound.

Preecho. Preechoes can best be understood by visualizing a stage environment, where the early reflections or preechoes are the sounds emanating from the rear and side stage walls directly after sound from the stage. Usually, the rear stage wall echo is earlier and louder than those from the two side walls.

The Preecho controls on the 224X change the perceived reflecting surfaces surrounding the source. Most programs in the 224X have four or more discrete preechoes, half panned left and half panned right. Each represents a reflecting surface and can be adjusted in level and time of delay using the respective sliders.

Changing amplitudes and delays of the preechoes changes the apparent geometrical arrangement of the acoustic environment. For example, to use Preecho controls to create an image of a rear wall, adjust the first two sliders to roughly the same delay time and amplitude. This keeps the images in the reflections panned as they were in the input to the 224X. If one of the two sliders is set earlier than the other by more than 2 milliseconds, the image tends to shift to that channel. If the difference in delay gets larger than a few milliseconds, the original panning of the source is lost, and the reverberation appears more spacious. (Delay differences of 2 milliseconds or so can cause problems when the reverberation is mixed to mono and have been avoided in most of the 224X program variations.)

The second pair of channels can be used to simulate side walls or the ceiling. These typically have lower amplitude and longer delays than the first pair of channels, and differences in their delays are less striking.

A stereo pair representing a rear stage wall could be set at a level of 30 with delay times between 20 to 30 milliseconds. By slightly varying the delay time between one channel and the other, a greater sense of space and dimension can be created. The preechoes from the side walls could be simulated by setting the delay times of a second stereo pair between 40 and 50 milliseconds with a greater offset between the two than the first pair. The level of this second pair would be set lower than the first pair.

A highly effective application of the preechoes is to fill in the space between the in-

ital sonic impulse and the delayed reverberation when a large value of predelay is used. The Rich Chamber program introduces diffusion to the preechoes, which makes them less obvious when they are used to create a sense of large size. Note that only the Rich Chamber program has this feature.

Gate. The Gate slider selects the amount of time delay introduced in switching between a program's running and stopped decay time. This control is only active when the Dynamic Decay toggle is turned on.

Size. The size control allows continuous adjustment of the acoustical "space" created by the 224X. Size is available on most programs and is program dependent in range of adjustment. Note that large size setting will limit the available pre-delay on some programs.

3.3.2 Parameter Toggles

The 224X has three parameter toggles that are accessed by successive pushes of the LARC's PARAM key. Once a toggle has been accessed, it can be turned on by pressing 1 on the numeric-select keypad, or off by pressing 0. The state of the parameter toggle is displayed in brackets after the parameter name; for example:

DYN DECAY [1]

indicates that dynamic decay is turned on. Parameter toggle settings are stored in registers along with other parameter settings. Note that most variations have toggles preset to on.

Dynamic Decay. This toggle activates the LF Stop Decay and Mid Stop Decay sliders. For an explanation of the function of these sliders, see the previous section "Dynamic Decay: LF and Mid Frequency Stop Decay."

Mode Enhancement. This toggle activates the Chorus slider. For an explanation of the function of this slider, see the previous section "Chorus."

Decay Optimization. This toggle activates a software module that alters certain reverberation parameters in response to changes in input level. These changes make the decay of sound in the 224X less metallic and more natural. This switch is normally on. However, for certain kinds of source material (e.g., soft low-frequency tones from a synthesizer) audible clicks may occur during level changes. If audible clicks occur, turn off the Decay Optimization toggle. For the Constant-Density Plate programs, this toggle has no function.

Errata

Model 224XL Owner's Manual

Pages 4.4 and 4.5

Maximum SIZE for the Concert Hall and Bright Hall programs is shown as 87 meters. It should be 40 meters.

Page 4.33

Maximum SIZE for the Plate/Chorus program is shown as 80 meters. It should be 70 meters.

Lexicon Part #070-04902

Programs

This section describes the reverberation and effects programs in the latest version of 224X software. The 224X's programs are organized into five banks:

- 1 Halls
- 2 Rooms
- 3 Plates
- 4 Effects
- 5 Splits

Sections 4.1 to 4.5 contain in-depth descriptions of the programs found in each bank, and Sec. 4.6 includes block diagrams of all programs.

Each bank comprises a family of several programs that have similar characteristics. Table 4.1 lists the programs supplied with the current version of 224X software.

Table 4.1. 224X Programs.

Programs	Banks				
	1 Halls	2 Rooms	3 Plates	4 Effects	5 Splits
1	Concert Hall 7 Variations	Room 4 Variations	Plate 6 Variations	Chorus & Echo 4 Variations	Hall/Hall 1 Variation
2	Bright Hall 5 Variations	Small Room 4 Variations	Small Plate 6 Variations	Resonant Chords 1 Variation	Plate/Plate 2 Variations
3	Dark Hall 7 Variations	Chamber 1 Variation	Constant Density Plate A 1 Variation	Multiband Delay 1 Variation	Plate/Hall 1 Variation
4		Rich Chamber 8 Variations	Constant Density Plate B 3 Variations		Plate/Chorus 1 Variation
5		Dark Chamber 8 Variations	Rich Plate 8 Variations		Rich Split 1 Variation
6		Inverse Room 3 Variations			

The Halls bank holds programs with a pronounced sense of large size and acoustic space. These programs lend space and ambience to recordings.

The Rooms bank holds the Room and Small Room programs, which also have a strong sense of space, but of smaller size than the Hall programs. The Room programs have a very wide range of uses in recording and broadcast. The Rooms bank also holds the Chamber programs, which have fewer size cues than the Halls.

The Plates bank holds programs that have high initial density and a smooth sizeless decay. These programs have a slight metallic tone and are widely used in mixing popular music.

In addition to reverberation programs, the 224X has several powerful special effects programs in the Effects bank that open up a whole new range of exciting possibilities traditionally provided only by other devices.

The Splits bank holds programs that allow the two input channels of the 224X to be processed independently, so entirely different reverberation sounds can be applied to different tracks in a mix.

Each program has one or more permanent variations. Each variation is a group of permanently set parameters that characterize the program for specific applications. Users can tailor these variations by changing the values of the parameters to suit their own applications and store the new parameter settings in registers for future use. Changing parameters is accomplished by accessing control pages, which are groupings of variable parameters whose values can be altered by moving the LARC's sliders. Most programs have four or five control pages, with each page holding as many as six variable parameters (corresponding to the LARC's six sliders). (Some pages have less than six variable parameters—for these cases, the unused sliders are inactive.)

In addition to the variable parameters, the 224X has three parameter toggles that are accessed through the PARAM key. The toggles are:

Dynamic Decay

Mode Enhancement

Decay Optimization.

All reverberation programs are preset with two parameter toggles on: Mode Enhancement and Decay Optimization (in the two Constant-Density Plate programs, the Decay Optimization toggle is inactive).

Adjustable Preecho Delays are preset for all reverberation programs, but the Preecho Levels are preset (nonzero) on only some variations. For more detail, see the descriptions of individual programs following in this section.

4.1

Halls—Bank 1

The reverberation from the three Hall programs (Concert, Bright, and Dark) is designed to sound as if it goes BEHIND the direct sound, adding ambience but leaving the source unchanged. These programs have a relatively low initial echo density, which gradually builds as time progresses. These programs emulate real concert halls, which accounts for their clean initial sound. The Hall programs are especially good with classical music. On popular music, they can give separately recorded tracks the sense of belonging to the same performance by putting the whole mix in the context of a real-sounding acoustic space.

The Concert and Bright Hall programs have both stereo or quad outputs. For stereo operation, use Output A for left and C for right. For quad, assign Outputs A and C to the left and right rear channels and B and D to the left and right front channels, respectively. Note: For quad, the Depth slider affects Outputs A and C only and should be set low (from 0 to 10).

The Dark Hall program has stereo outputs: the left output is A and the right output is C.

4.1.1 Concert Hall—Program 1

Program 1, Concert Hall, was previously called the Main Hall (Program 1) in the 224. It emulates a concert hall about 120 feet long. The Treble Decay slider filters all the sound except the preechoes, giving a darker tone to the reverb than the Bright Hall program (program 2). This darker tone simulates

the effect of air absorption in a real hall and helps keep the ambience generated by the program from muddying the direct sound. Additional treble rolloff can be added by lowering the HF Bandwidth slider, which also darkens the Preecho Delays.

Variations in Concert Hall. The Concert Hall program has five control pages and seven variations. Table 4.2 lists the control pages and the variable parameters and their ranges. All variations have four adjustable Preecho Delays; variations 2, 4, and 6 have active Preechoes.

Variation 1 mimics a moderately large, reverberant hall. Average running reverb decay time is 2.6 seconds: LF Decay (3.0 seconds) is longer than Mid Decay (2.0 seconds). Diffusion is 25, and Depth is moderately low (33). Predelay is 24.0 milliseconds—the minimum for this program.

Variation 2 sounds especially good with classical music. It emulates a smaller hall sound than variation 1; both LF and Mid Decay are 1.7 seconds. Diffusion is lower (15) than in variation 1, and there are four Preechoes. Levels are preset to add a sense of stage reflections not present in a closely miked track.

Variation 3 has a very low Diffusion (01) that, along with its fairly long initial reverb decay time, produces an uncolored, natural quality on flute or voice. This variation works especially well when modified to produce very long reverb times; it is not for material containing strong transients. This variation was previously called Large Concert Hall (program 3) in the 224.

Variation 4 is a modification of variation 3 with a slightly longer Mid Decay (3.0 seconds), less HF Bandwidth (7.5 kHz instead of 9.0), longer Predelay (42.0 milliseconds), and four active Preechoes.

Variation 5 is just like variation 1, except that the average reverb decay time has been increased to 6.5 seconds.

Variation 6 is preset with the Dynamic Decay toggle on; average stopped reverb decay time (5.7 seconds) is longer than average running reverb decay (1.7 seconds), and it contains four preset Preecho Levels. This configuration creates a wash of reverb when the music or input source stops, but maintains clarity at other times.

Variation 7 is also preset with the Dynamic Decay toggle on, but with the variable reverb parameter values exactly opposite to those in variation 6; i.e., the average running decay time (5.7 seconds) is longer than the average stopped decay (1.7 seconds). The result is similar to what might be obtained if the output of a reverberation device were put through a gating circuit.

4.1.2 Bright Hall—Program 2

Program 2, Bright Hall, is similar to the Concert Hall program, except the Treble Decay slider affects the sound only after a few hundred milliseconds. The sound from this program is thus much brighter than that from the Concert Hall program, and many people prefer this brightness in popular music. Diffusion has also been set higher in all the variations, enhancing percussion sounds. If an even brighter sound is wanted, turn off the Mode Enhancement toggle.

Variations in Bright Hall. This program has five control pages and five variations. Table 4.3 lists the control pages and the variable parameter ranges. All variations have four adjustable preecho delays; variations 2 and 4 have active Preechoes.

For variations 1, 3, and 5, the HF Bandwidth control is preset to 19 kHz; for variation 2, it is 9.0 kHz; and for variation 4, 7.50 kHz.

Table 4.2. Concert Hall—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Depth	Predelay
1	70 sec	70 sec	19.0 kHz	19.0 kHz	99	216 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	24.0 msec
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
2	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
	Preecho Level 1 L>A	Preecho Level 2 R>C	Preecho Level 3 R>A	Preecho Level 4 L>C		
3	99	99	99	99	Inactive	Inactive
	00	00	00	00		
	Preecho Delay 1 L>A	Preecho Delay 2 R>C	Preecho Delay 3 R>A	Preecho Delay 4 L>C	Fine Predelay L>	Fine Predelay R>
4*	188 msec	188 msec	188 msec	188 msec	31.3 msec	31.3 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Size		Gate			
5	87		5.08 sec.			
	08		0.00 sec.			
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

Table 4.3. Bright Hall—Control
Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Depth	Predelay
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	216 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	24.0 msec
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
3	Preecho Level 1 L>A	Preecho Level 2 R>C	Preecho Level 3 R>A	Preecho Level 4 L>C	Inactive	Inactive
	99	99	99	99		
	00	00	00	00		
4*	Preecho Delay 1 L>A	Preecho Delay 2 R>C	Preecho Delay 3 R>A	Preecho Delay 4 L>C	Fine Predelay L>	Fine Predelay R>
	188 msec	188 msec	188 msec	188 msec	31.3 msec	31.3 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
5	Size		Gate			
	87		5.08 sec			
	08 meters		0.00 sec.			
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

4.1.3 Dark Hall—Program 3

Program 3, Dark Hall, is noticeably less metallic and more natural sounding than the other hall programs, especially as the sound decays. A side effect of this is a markedly darker tone color. The darkening can be removed by turning the Mode Enhancement Toggle off, which makes the reverb much brighter and more metallic. Additionally, the Treble Decay slider, which affects the sound only after a few hundred milliseconds, can be raised to compensate for the increased darkness. The Dark Hall program is the best choice when a hall sound is needed in classical music, or, for that matter, any time its darker color can be used to advantage.

Note: *The Chorus control is preset to 53 in all variations of the Dark Hall program. If pitch wobble is heard in material with very pure tones, such as piano or guitar, set the Chorus control to 50 or lower. Lowering this control maintains the dark tone color, but makes the sound more metallic.*

Variations in Dark Hall. This program has five control pages and seven variations. Table 4.4 lists the control pages and variable parameter ranges. All variations have four adjustable Preecho Delays; variations 2, 4, and 6 contain active Preechoes. Note that these variations are similar in sound to those in the Concert Hall program.

Table 4.4. Dark Hall—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Depth	Predelay
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	176 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	24.0 msec
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
3	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB		
	99	99	99	99	Inactive	Inactive
	00	00	00	00		
4*	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Fine Predelay L>	Fine Predelay R>
	143 msec	143 msec	143 msec	143 msec	31.3 msec	31.3 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
5	Size		Gate			
	40		5.08 sec.			
	05		0.00 sec.			
	1	2	3	4	5	6
	Sliders					

* The software for this page allows fine tuning.

4.2

Rooms—Bank 2

This bank comprises the Room, Small Room, Chamber, Rich Chamber, Dark Chamber, and Inverse Room programs. The general configuration of the control pages for the Room programs is the same as for the Hall programs; the Chamber and Inverse Room programs are slightly different.

Like the Concert and Bright Hall programs, the Room program has both stereo or quad outputs. For stereo operation, use Output A for left and C for right. For quad, assign Outputs A and C to the left and right rear channels and B and D to the left and right front channels, respectively. Note: The Depth slider affects Outputs A and C only.

The Small Room, Rich and Dark Chamber programs have stereo outputs: the left output is A and the right is C.

The Chamber program also has stereo outputs; however, this program digitally averages its two inputs to mono. The main left output is A and the main right output is C. If only one input feed is available, bridge it to both Right and Left inputs. Note that Output A has less inherent predelay than Output C, and this difference is noticeable on some material as a shift in apparent position in the stereo image. Outputs D and B, which are derived from A and C and do not have different predelays, can be used for better timing characteristics but introduce a marked coloration from Output D. For a single (mono) output, do not use Output D because of its coloration.

4.2.1 Room and Small Room—Programs 1 and 2

Programs 1 and 2, Room and Small Room, are similar to the Hall programs, but the spaces they emulate are smaller. The Room programs have higher effective diffusion than the halls, because sound evens out more quickly in smaller spaces. They are useful when density or a sense of acoustic space needs to be added to a sound. The Room program emulates a space 30 to 50 feet long (about 1/2 the size and 1/8 the volume of the Concert Hall program), and Small Room a space about half that length (1/8th the volume again). The Small Room program in particular increases the apparent loudness of spoken material without raising its peak level or degrading intelligibility. (The reverberation has high articulation.) Both programs are ideal for broadcast or film work and can be very useful on vocals or drums.

The Small Room program is ideal for dubbing film dialog in scenes that take place in confined spaces. In addition, it is very good for fattening vocals and for enhancing percussion. Like the Dark Hall program, the Small Room program is noticeably less metallic and more

natural sounding, especially as the sound decays, which darkens the tone of the reverb. To some degree, this darkening can be compensated for by raising the Treble Decay slider, or it can be removed entirely by turning off the Mode Enhancement toggle, which makes the reverb much brighter and more metallic.

Note: *The Chorus control is preset to 53 in all variations of the Room and Small Room programs. If pitch wobble is heard in material with very pure tones, such as piano or guitar, set the Chorus control to 50 or lower. Lowering this control maintains the dark tone color, but makes the sound more metallic.*

Variations in Room and Small Room. The Room program has five control pages and four variations. Table 4.5 lists the control pages and variable parameter ranges. All variations have four adjustable Preecho Delays, variations 2 and 4 have four active Preechoes.

Table 4.5. Room—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Depth	Predelay
1	35 sec	35 sec	19.0 kHz	19.0 kHz	99	360 msec
	0.3 sec	0.3 sec	170 Hz	170 Hz	00	24.0 msec
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
2	35 sec	35 sec	97	19.0 kHz	99	75
	0.3 sec	0.3 sec	00	170 Hz	00	00
	Preecho Level 1 L>A	Preecho Level 2 R>C	Preecho Level 3 R>A	Preecho Level 4 L>C		
3	99	99	99	99	Inactive	Inactive
	00	00	00	00		
	Preecho Delay 1 L>A	Preecho Delay 2 R>C	Preecho Delay 3 R>A	Preecho Delay 4 L>C	Fine Predelay L>	Fine Predelay R>
4*	322 msec	322 msec	322 msec	322 msec	30.2 msec	30.2 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Size		Gate			
5	87 meters		1.26 sec			
	08 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

Variation 1 has a fairly long average reverb decay time (1.4 seconds) given the small size of the space that it emulates. The HF Bandwidth control is preset at 7.5 kHz, and Depth is fairly high at 33.

Variation 2 is similar to variation 1, except that the four Preecho Delays are preset at moderate levels, ranging from 6.5 to 21 milliseconds.

Variation 3 has a short average running reverb decay time (0.5 seconds) with full HF Bandwidth (19.0 kHz), but with rapid Treble Decay (4.5 kHz). This variation is especially suitable for percussion.

Variation 4 is a modified version of variation 3 and has four Preecho Delays preset to much higher levels than in variation 2.

Table 4.6. Small Room—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Depth	Predelay
	23 sec	23 sec	19.0 kHz	19.0 kHz	99	348 msec
	0.2 sec	0.2 sec	170 Hz	170 Hz	00	12.0 msec
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
	23 sec	23 sec	97	19.0 kHz	99	75
	0.2 sec	0.2 sec	00	170 Hz	00	00
3	Preecho Level 1 L > AD	Preecho Level 2 R > CB	Preecho Level 3 R > AD	Preecho Level 4 L > CB		
	99	99	99	99	Inactive	Inactive
	00	00	00	00		
4*	Preecho Delay 1 L > AD	Preecho Delay 2 R > CB	Preecho Delay 3 R > AD	Preecho Delay 4 L > CB	Fine Predelay L >	Fine Predelay R >
	322 msec	322 msec	322 msec	322 msec	30.2 msec	30.2 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	1	2	3	4	5	6
	Sliders					

The software for this page allows fine tuning.

The Small Room program has four control pages and four variations that include very short reverb decay times and high articulation. Table 4.6 lists the control pages and variable parameter ranges. Variations 2 and 4

have four preset Preecho Delays that are similar to those in the Room program, except that both the decay times and predelay times are shorter.

4.2.2 Chamber, Rich Chamber, and Dark Chamber Programs 3, 4, and 5

Programs 3, 4, and 5, Chamber, Rich Chamber, and Dark Chamber, have some attributes of both the Hall and Plate programs (see Sec. 4.3). They have very few size cues—giving a relatively smooth decay with time after a short build-up period.

The Chamber program has relatively low initial diffusion, even with the Diffusion control raised; however, diffusion increases rapidly after a few hundred milliseconds. This program sounds like a small echo chamber, but with less initial diffusion. It works well on many types of popular music, and sounds very different from the Plate programs because of its initial sound. The Depth control found in the Hall and Room programs is replaced by an Attack control on page 1, which controls the explosiveness of attack on percussion.

Caution: *The Chamber program may feed back internally if the Mid Decay is set much higher than the LF Decay and if the Treble Decay and HF Bandwidth sliders are set too high. This feedback can be defeated by lowering the Treble Decay slider (it might, of course, be useful as a special effect).*

The algorithm used in Rich Chamber produces an even, relatively dimensionless reverberation, with little change in color as the sound decays. The initial diffusion is similar to the Hall or Room programs, but the sense of space or size is much less obvious. This characteristic, along with the low color in the decay tail, make the Rich Chamber program useful for a wide variety of material.

When the Diffusion control is set to a low or moderate level, this program is good on classical music, especially piano (where a short reverberation time is recommended) or organ music (with long reverberation times). With a

high diffusion setting, the program emulates a well-diffused large acoustic chamber and is exciting on all types of popular music.

The Dark Chamber is very similar to the Rich Chamber. The primary difference is that the Dark Chamber has a sharp filter which limits its response above 10 kHz. This emulates the effect of air absorption in a real acoustic space, providing a very natural sound. The Dark Chamber is useful in a wide variety of classical and mixed popular music.

The Rich and Dark Chambers have six adjustable Preecho Delays, but unlike most 224X programs, these Preecho Delays are affected by the Diffusion control. As the Diffusion slider is raised, each preecho becomes a diffused cluster. This added diffusion allows the Preecho Delays to be used to create an adjustable sense of space to the otherwise dimensionless reverberation. For example, a large hall sound could be created by setting the Predelay control to 100 milliseconds or so, and then filling in the sound before this delay with diffused preechoes. Clusters of preechoes of various amplitudes around 30 and 75 milliseconds seem to be quite effective in putting the basic reverberation behind the music, not on top of it. Even with such preecho and predelay manipulation, the sense of spaciousness is not as great as in the Hall programs, which are probably preferable when a great increase in spaciousness is wanted. However, the Rich and Dark Chamber programs seem to provide a greater increase in apparent loudness and richness for the same peak level than the Hall programs and has lower color, especially in the decay tail. The Rich and Dark Chamber programs are excellent on spoken voice, giving a good increase in loudness with very low color.

Table 4.7. Chamber—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
1	70 sec	70 sec	19.0 kHz	19.0 kHz	99	249 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	25.0 msec
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	
2	70 sec	70 sec	97	19.0 kHz	99	Inactive
	0.6 sec	0.6 sec	00	170 Hz	00	
	1	2	3	4	5	6
	Sliders					

The Size control adds a great deal of flexibility to the Rich Chamber program. When set to sizes of 16 meters or less, the Rich Chamber is tight and articulate—a very useful sound for dialog and music. Larger sizes (around 60 meters) give an excellent concert hall sound, especially if the Diffusion control is set to about 50.

Low settings of Treble Decay can cause some unevenness in decay with these programs. Settings below about 6 kHz with long reverb times can be problematic. If a darker sound is wanted, use the HF Bandwidth control, not the Treble Decay control.

Variations. The Chamber program has two control pages and only one variation. Table

4.7 lists its control pages and variable parameter ranges.

The Rich Chamber and Dark Chamber programs have five control pages and eight variations each. Tables 4.7 and 4.8 list the control pages and variable parameter ranges for the Rich Chamber and Dark Chamber, respectively. All variations have six adjustable Preecho Delays that are affected by the diffusion slider. Variations 2, 3, 5, and 6 have active Preechoes and emulate spaces of increasing size. Variation 4 is a medium size, high diffusion room, useful for percussion. Variation 7 is preset to provide a demonstration of the Infinite Reverb control feature. Variation 8 is designed to be used for Gated Reverb effects.

Table 4.8. Rich Chamber—Control
Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
	83 sec*	83 sec*	19.0 kHz*	19.0 kHz*	99	834 ms
	0.1 sec	0.1 sec	170 Hz	170 Hz	00	0.00 ms
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
	83 sec*	83 sec*	99	19 kHz*	99	99
	0.1 sec	0.1 sec	00	170 Hz	00	00
3	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
	99	99	99	99	99	99
	00	00	00	00	00	00
4	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
	125 ms	125 ms	125 ms	125 ms	125 ms	125 ms
	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms
5	Size	Inactive	Reverb Stop Delay (Gate)	Inactive	Inactive	Inactive
	87 meters		1.26 sec			
	08 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

* Can also be set to infinite.

Table 4.9. Dark Chamber—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
	83 sec *	83 sec *	19.0 kHz *	19.0 kHz *	99	830 ms
	0.1 sec	0.1 sec	170 Hz	170 Hz	00	000 ms
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
	83 sec *	83 sec *	99	19.0 kHz *	99	99
	0.1 sec	0.1 sec	00	170 Hz	00	00
3	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
	99	99	99	99	99	99
	00	00	00	00	00	00
4	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
	125 ms	125 ms	125 ms	125 ms	125 ms	125 ms
	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms
5	Size	Inactive	Reverb Stop Delay (Gate)	Inactive	Inactive	Inactive
	87 meters		1.26 sec			
	08 meters		0.00 sec			
	1	2	3	4	5	6
Sliders						

*Can also be set to infinite.

Table 4.10. Inverse Room—Control
Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
1	83 sec	83 sec	19 kHz	19. kHz	99	830 ms
	0.1 sec	0.1 sec	170 Hz	170 Hz	00	000 ms
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
2	83 sec	83 sec	99	19 kHz	99	99
	0.1 sec	0.1 sec	00	170 Hz	00	00
	Decay Slope	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
3	99	99	99	99	99	99
	00	00	00	00	00	00
	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
4	125 ms	125 ms	125 ms	125 ms	125 ms	125 ms
	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms
	Size 1		Reverb Stop Delay (Gate)			
5	87 meters		1.26 sec			
	08 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

4.2.3 Inverse Room — Program 6

The Inverse Room program allows the envelope of the reverb tail to be continuously varied, using the Slope control. The effect is similar to a gate, but does not depend at all on the level or complexity of the input signal.

When the Slope control is set at less than 50, the Inverse Room program produces a reverb that sounds similar to a normal room for a fraction of a second, and then drops off abruptly. The length of time until the sound cuts off is set by the Size control. The apparent reverb time until the sound abruptly ends is set by the Slope control.

If the Slope control is set at 50, the sound does not decay until the cut-off (i.e., amplitude is constant). This is sometimes referred to as a level slope effect.

If the Slope control is set above 50, the sound increases in level until the cutoff, producing what is sometimes referred to as inverse reverb. The resulting sound can have great impact and interest.

The Inverse Room program can also be used to enhance a vocal or speaking voice by adding volume without adding apparent reverb or increasing the peak level of the input signal. To produce this "enhance" effect, set the Slope control to about 30, and the diffusion control to about 20. Use the Size control to match the articulation of the input signal. For speech, size values of 10 to 20 meters are a good starting point.

Variations. The Inverse Room program has five control pages and three variations. Table 4.10 lists the control pages and variable parameter ranges.

Variation 1 produces a level slope effect, variation 2 produces inverse reverb, and Variation 3 is preset for the enhance effect.

The Plate programs have high initial diffusion and bright, colored sound. For this reason, they have traditionally been chosen for percussion. With the variable parameters available on the 224X, they are useful for a wider variety of tasks as well. The 224X has five Plate programs: Plate, Small Plate,

Constant-Density (CD) Plate A, Constant-Density (CD) Plate B, and Rich Plate. The Plate programs have four or five control pages; an Attack slider on page 1 controls the explosiveness of the attack on percussive material.

4.3.1 Plate and Small Plate—Programs 1 and 2

The Plate program mimics the sounds of many types of metal plates and was the original plate program in the 224. When the Diffusion control is set low, the Plate program has a very clear sound that is excellent on vocals and can be used with Preechoes to create a wide variety of acoustic environments. When diffusion is high, this program gives a smooth, dense sound with applications in all popular music. Mono compatibility of the outputs is very good, but for best results percussive material should be panned to the middle of the stereo feed.

The Small Plate program is almost an exact duplicate of the Plate program, except that it sounds tighter, more diffuse, and smoother, especially on transients. Its very high diffusion gives it a characteristically mellow sound that is useful on a wide variety of popular music, especially percussion.

Variations in Plate and Small Plate. The Plate program has five control pages and six variations. Table 4.11 lists the control pages and the variable parameter ranges. All variations have six preset Preecho Delays; variations 3 and 4 have active Preechoes.

Variation 1 is brighter and more metallic-sounding than any of the Hall programs. Attack and Predelay are both 0, giving a crisp, slightly thin sound, and a strong attack. Diffusion is high (58). Average running reverb decay time is 1.8 seconds, and the HF Bandwidth control is preset at 7.5 kHz.

Variation 2's short average reverb decay time (0.6 seconds for low frequencies and 1.8 seconds for midrange frequencies) and very low crossover (170 Hz) create a hard, slightly garagelike quality. With drums, the sound suggests a small hard-walled space without booming from the kick drum. The HF Bandwidth control is set to maximum—19 kHz.

Variation 3 has a reduced HF Bandwidth setting (7.50 kHz) to mimic air absorption. Attack is high (80) to eliminate explosive sound, Diffusion is low (31), Definition is high (58), and there is long reverb decay time (4.2 seconds for low frequencies and 3.0 seconds for midrange frequencies). This variation sounds like a large, stone-lined church. It has six Preechoes, creating the impression of an actual church acoustical environment instead of simply adding a "churchy" sound to a dry track.

Variation 4 is a version of variation 3, with longer average reverb decay time: 7.5 seconds for low frequencies and 5.2 seconds for midrange frequencies. Like variation 3, it has six Preechoes. This variation produces a cavernous quality typical of large reverberant spaces.

Variation 5 is an updated version of the percussion plate program that was called program 5 in the 224. Its very high initial diffusion (58) can be decreased to make a clear sound for vocals.

Table 4.11. Plate — Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
1	70 sec	70 sec	19.0 kHz	19.0 kHz	99	176 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
2	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
3	00	00	00	00	00	00
	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
	170 msec	170 msec	170 msec	170 msec	170 msec	170 msec
4*	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Size		Gate			
	40 meters		5.08 sec			
5	10 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

Variation 6 is like variation 3, but without Preechoes. It has a long reverb decay time (4.2 seconds for low frequencies and 3.0 seconds for midrange frequencies), but leaves more space around the performer. Although it does not simulate a church as realistically as variation 3, it is useful for adding churchlike reverb to recordings that already have some spaciousness, such as organ recordings made in moderately dry spaces.

Like the Plate program, the Small Plate program has five control pages with six variations. Table 4.12 lists the control pages and the variable parameter ranges. All variations have six preset Preecho Delays; variations 3 and 4 have active Preechoes.

Table 4.12. Small Plate—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
1	44 sec	44 sec	19.0 kHz	19.0 kHz	99	304 msec
	0.4 sec	0.4 sec	170 Hz	170 Hz	00	0.00 msec
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
2	44 sec	44 sec	97	19.0 kHz	99	75
	0.4 sec	0.4 sec	00	170 Hz	00	00
	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
3	99	99	99	99	99	99
	00	00	00	00	00	00
	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
4*	286 msec	286 msec	286 msec	286 msec	286 msec	286 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Size		Gate			
5	40 meters		5.08 sec			
	10 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

4.3.2 Constant-Density (CD) Plates A and B—Programs 3 and 4

The density of the echoes in natural acoustic spaces, as in most 224X programs, increases with time. When a sound is made in a hall, relatively few reflections exist at first, but as the number of echoes increases, so do both the smoothness of the reverb and the amount of coloration in the sound. The CD Plate programs act differently. They start out with very high initial diffusion and maintain a constant echo density thereafter. The rate of

decay is also constant with time, instead of beginning rapidly and then slowing as time progresses. For these programs, the Decay Optimization toggle is inactive.

The CD Plate A program was originally developed for the 224; it emulates a sound that is well known in the industry; CD Plate B has a more spacious sound and better mono compatibility.

Table 4.13. Constant-Density Plate A—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	181 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	5.00 msec
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	
	70 sec	70 sec	56	19.0 kHz	99	Inactive
	0.6 sec	0.6 sec	00	170 Hz	00	
3	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
	99	99	99	99	99	99
	00	00	00	00	00	00
4*	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
	170 msec	170 msec	170 msec	170 msec	170 msec	170 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

Variations in Constant-Density Plates A and B. The CD Plate A program has four control pages and only one variation. The variation has six active Preechoes. Table 4.13 lists the control pages and variable parameter ranges. Average running reverb decay time is 1.8 seconds, with an HF Bandwidth of 19.0 kHz and a Treble Decay of 15 kHz. Attack is preset at 12, and Diffusion at 58.

The CD Plate B program has four control pages and three variations. Table 4.14 lists the control pages and the variable parameter ranges. All variations have six preset preecho delays; variation 3 has active Preechoes.

Table 4.14. Constant-Density Plate
B—Control Pages and Variable
Parameters.

Page	Variable Parameters and Ranges					
	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
1	70 sec	70 sec	19.0 kHz	19.0 kHz	99	120 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	
2	70 sec	70 sec	56	19.0 kHz	99	Inactive
	0.6 sec	0.6 sec	00	170 Hz	00	
	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
3	99	99	99	99	99	99
	00	00	00	00	00	00
	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
4*	121 msec	121 msec	121 msec	121 msec	121 msec	121 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

Variation 1 is much like CD Plate A. Its average running reverb decay time is 1.8 seconds, HF Bandwidth is preset at 7.5 kHz, and Treble Decay at 15 kHz. Attack is preset at 33 and Diffusion at 58.

Variation 2 produces a brighter sound, with short reverb decay time, especially suited to percussion. The HF Bandwidth control is a full 19 kHz, and LF and Mid Decay are both at 0.6 seconds.

Variation 3 is like variation 1, but with four Preechoes, all occurring before 10 milliseconds.

Table 4.15. Rich Plate—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay	Mid Decay	Crossover	Treble Decay	Attack	Predelay
	79 sec*	79 sec*	19.0 kHz*	19.0 kHz*	99	838 ms
	0.1 sec	0.1 sec	170 Hz	170 Hz	00	000 ms
2	LF Stop Decay	Mid Stop Decay	Chorus	HF Bandwidth	Diffusion	Definition
	79 sec*	79 sec*	97	19 kHz*	99	99
	0.1 sec	0.1 sec	00	170 Hz	00	00
3	Preecho Level 1 L>AD	Preecho Level 2 R>CB	Preecho Level 3 R>AD	Preecho Level 4 L>CB	Preecho Level 5 L>AD	Preecho Level 6 R>CB
	99	99	99	99	99	99
	00	00	00	00	00	00
4	Preecho Delay 1 L>AD	Preecho Delay 2 R>CB	Preecho Delay 3 R>AD	Preecho Delay 4 L>CB	Preecho Delay 5 L>AD	Preecho Delay 6 R>CB
	125 ms	125 ms	125 ms	125 ms	125 ms	125 ms
	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms
5	Size	Inactive	Reverb Stop Delay (Gate)	Inactive	Inactive	Inactive
	83 meters		1.26 sec			
	08 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

*Can also be set to infinite.

4.3.3 Rich Plate — Program 5

The Rich Plate Program is denser, smoother, and less colored than the other plate programs. When used with the Size control set at around 16 meters, the sound is dense and tight—ideal for percussion. Larger sizes and longer reverb times are suitable for vocals and brass.

The Rich Plate program has five control pages and eight variations. Table 4.15 lists

the control pages and variable parameter ranges.

Variations 1 through 3 have moderate size with increasing preechoes, and are useful for vocals and mixed music. Variations 4 through 6 are tighter and denser, well-suited to percussion. Variation 7 is preset to produce infinite reverb, and Variation 8 produces a gated reverb effect.

The three Effects programs (Chorus & Echo, Resonant Chords, and Multiband Delay) are unique and were designed for special audio effects and modifications. These programs

offer many exciting possibilities that were previously unavailable on digital reverberators. All Effects programs and their variations have stereo outputs.

4.4.1 Chorus & Echo—Program 1

The Chorus & Echo program generates six voices, three on each input channel. This program provides many different effects. By adjusting the delays, the amount of feedback, and the panning of each voice, doubling, tripling, flanging, echo flanging, and other sounds can be created.

The Chorus & Echo program has five control pages and four variations. Table 4.16 lists the control pages and the variable parameter ranges. The control page configuration differs from those in the reverberation programs. Essentially, the first page of controls affects the six voices in unison, and the remaining five pages control individual voices from the Right and Left inputs: the first three sliders affect signals from the Left input and the last three sliders affect signals from the Right.

Page 1 has three active controls. Sliders 1, 2, and 6 are inactive; slider 3 controls Chorus, which varies the intensity of the random delay variation (pitch shifting) to help keep the voices separate. Slider 4 controls high-frequency bandwidth, which (instead of being a simple input filter) affects all channels and the recirculated sound as well. Slider 5 controls diffusion on all channels.

Page 2 controls each Voice Level (strength of each voice), and page 3 varies each voice delay (amount of time before voice appears at output).

Page 4 controls the gain of voice feedback, which determines the length of resonance. The controls are zero-center, with positive feedback above and negative below.

Page 5 assigns the voices to the two output channels. As the control is moved from bottom to top, the voices pan from Outputs C and B to Outputs A and D.

The four variations in the Chorus & Echo program demonstrate some of the amazing possibilities available. Variations 1 and 2 have normal six-voice doubling, both with medium depth; however, variation 1 has faster vibrato than variation 2.

Note: Fast vibrato is unique to variation 1 and is not variable by the user. This variation is the only one in the 224X that cannot be made exactly like the other variations that share its program.

Variation 3 is a six-voice echo chorus with flanging and very strong pitch-shifting. In addition, almost a half-second delay occurs in the arrival of the first voice. Pitch-shifting can be moderated while retaining the flanging effects by moving slider 3 on page 1 down to about 50. The feedback can be reduced by bringing up sliders 1 and 4 on page 4 to near the center of their travel.

Variation 4 is like variation 3, except that the built-in delays are much shorter, and the initial values on the feedback page (page 4) are lower.

Note: Variations 3 and 4 use a great deal of feedback to create their sound, which raises the internal signal level in the processor. For these variations, the 224X's processor overloads well before the incoming level reaches +12 dB. Watch the LARC's overload (ovld) LEDs carefully, and reduce the input level if overload occurs.

Table 4.16. Chorus & Echo—Control
Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	97	19.0 kHz	99	
			00	170 Hz	00	
	Voice Level 1 L>	Voice Level 2 L>	Voice Level 3 L>	Voice Level 4 R>	Voice Level 5 R>	Voice Level 6 R>
2	99	99	99	99	99	99
	00	00	00	00	00	00
	Voice Delay 1 L>	Voice Delay 2 L>	Voice Delay 3 L>	Voice Delay 4 R>	Voice Delay 5 R>	Voice Delay 6 R>
3*	896 msec	896 msec	896 msec	896 msec	896 msec	896 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Feedback 1 L>	Feedback 2 L>	Feedback 3 L>	Feedback 4 R>	Feedback 5 R>	Feedback 6 R>
4	97%	97%	97%	97%	97%	97%
	–97%	–97%	–97%	–97%	–97%	–97%
	Pan 1 CB-AD	Pan 2 CB-AD	Pan 3 CB-AD	Pan 4 CB-AD	Pan 5 CB-AD	Pan 6 CB-AD
5	99	99	99	99	99	99
	00	00	00	00	00	00
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

4.4.2 Resonant Chords—Program 2

The Resonant Chords program is unlike anything that has ever appeared in a delay device. It uses impulsive energy at the input to excite six resonant notes. The level, pitch, duration of ringing, and high-frequency rolloff of the overtones of each note are all separately controllable.

The notes resonate to some degree with almost any input, but the most effective excitation contains all frequencies, like percussion. A fairly simple drum track works, as does a timed pulse, such as a metronome. Other instruments, such as guitars, may give a quality of tonal ambience in which certain notes rise out of the background in an ethereal way.

This program has six control pages and only one variation. Table 4.17 lists the control pages and the variable parameter ranges.

The control page configuration differs from those in the reverberation programs; the six pages control individual voices from the Right and Left inputs: the first three sliders affect signals from the Left input and the last three sliders affect signals from the Right.

Page 1 controls level, which determines the strength of each note.

Page 2 determines the frequency of the tone. The shorter the delay, the higher the frequency at which the feedback loop resonates: THE PITCH GOES DOWN AS THE SLIDER IS MOVED UPWARD.

Page 3 controls feedback, which determines how long the note resonates. The controls are zero-center, with positive feedback above and negative below. Positive feedback generates all the harmonics of each note, and negative feedback gives only the odd harmonics. The action of these sliders takes place almost entirely in the last 20% of their travel, where the magnitude of the feedback coefficient is between 85 and 97%. Lower values do not produce separately audible notes, but can be used for subtle equalization effects.

Page 4 controls predelay, which sets the timing of the notes. By gradually increasing the delay with each note, a pulse can become a strum or the notes can be synchronized with the internal rhythms of the drum track.

Page 5 assigns the notes to the two output channels. As the sliders are moved from bottom to top, the sound pans from Outputs C and B to Outputs A and D.

Page 6 has two functions: (1) slider 1 feeds the output of Predelay 3 back to the Left input and the output of the Predelay 6 back to the Right input; in addition, slider 2 also controls feedback from 3 and 6 Predelays, but reverses them, so feedback from Predelay 6 goes to the left input and feedback from Predelay 3 goes to the right input (see Fig. 4.Z), and (2) sliders 3 and 4 control HF Bandwidth for the recirculation associated with the Left and Right inputs, respectively.

4.4.3 Multiband Delay—Program 3

The Multiband Delay program provides six separately adjustable delay outputs, each with its own high- and low-cut filters. It is a monaural input program, the inputs being digitally averaged.

This program has one variation with six control pages. Table 4.18 lists the control pages and the variable parameter ranges.

Page 1 adjusts the levels of individual bands.

Page 2 controls the delay. Short delays are good for slap-echo effects, but very long delays are also available for harmonizing and layering in real time.

Page 3 controls the low-frequency band edges of the high-pass filters, and page 4 controls the high-frequency band edges of the low-pass filters. These filters have a rolloff of 6 db/octave.

Page 5 assigns the bands to the two output channels. As the sliders are moved from bottom to top, the sound pans from Outputs C and B to Outputs A and D.

Page 6 is the overall feedback page. Sliders 1 and 2 feed the outputs of the corresponding delays back to the input summing junction. Slider 5 controls diffusion at the input of the program; raising the diffusion spreads transients out in time, turning a click into a brush stroke. Sliders 3, 4, and 6 are inactive.

Programs

Table 4.17. Resonant Chords—
Control Pages and Variable
Parameters.

Page	Variable Parameters and Ranges					
1	Note Level 1 L>	Note Level 2 L>	Note Level 3 L>	Note Level 4 R>	Note Level 5 R>	Note Level 6 R>
	99	99	99	99	99	99
2*	Note Pitch 1 L>	Note Pitch 2 L>	Note Pitch 3 L>	Note Pitch 4 R>	Note Pitch 5 R>	Note Pitch 6 R>
	31.3 msec	31.3 msec	31.3 msec	31.3 msec	31.3 msec	31.3 msec
3	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Resonance 1 L>	Resonance 2 L>	Resonance 3 L>	Resonance 4 R>	Resonance 5 R>	Resonance 6 R>
4	97%	97%	97%	97%	97%	97%
	–97%	–97%	–97%	–97%	–97%	–97%
5	Predelay 1 L>	Predelay 2 L>	Predelay 3 L>	Predelay 4 R>	Predelay 5 R>	Predelay 6 R>
	864 msec	864 msec	864 msec	864 msec	864 msec	864 msec
6	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Pan 1 CB-AD	Pan 2 CB-AD	Pan 3 CB-AD	Pan 4 CB-AD	Pan 5 CB-AD	Pan 6 CB-AD
7	99	99	99	99	99	99
	00	00	00	00	00	00
8	Crossfeed	Crossfeed	HF Cutoff L	HF Cutoff R		
	99%	99%	19.0 kHz	19.0 kHz	Inactive	Inactive
9	00	00	170 Hz	170 Hz		
	1	2	3	4	5	6
Sliders						

* The software for this page allows fine tuning.

Table 4.18. Multiband Delay—
Control Pages and Variable
Parameters.

Page	Variable Parameters and Ranges					
	Band Level 1 L+R>	Band Level 2 L+R>	Band Level 3 L+R>	Band Level 4 L+R>	Band Level 5 L+R>	Band Level 6 L+R>
1	99	99	99	99	99	99
	00	00	00	00	00	00
	Band Delay 1 L+R>	Band Delay 2 L+R>	Band Delay 3 L+R>	Band Delay 4 L+R>	Band Delay 5 L+R>	Band Delay 6 L+R>
2*	1.86 sec	1.86 sec	1.86 sec	1.86 sec	1.86 sec	1.86 sec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	LF Cutoff 1 L+R>	LF Cutoff 2 L+R>	LF Cutoff 3 L+R>	LF Cutoff 4 L+R>	LF Cutoff 5 L+R>	LF Cutoff 6 L+R>
3	19.0 kHz	19.0 kHz	19.0 kHz	19.0 kHz	19.0 kHz	19.0 kHz
	170 Hz	170 Hz	170 Hz	170 Hz	170 Hz	170 Hz
	HF Cutoff 1 L+R>	HF Cutoff 2 L+R>	HF Cutoff 3 L+R>	HF Cutoff 4 L+R>	HF Cutoff 5 L+R>	HF Cutoff 6 L+R>
4	19.0 kHz	19.0 kHz	19.0 kHz	19.0 kHz	19.0 kHz	19.0 kHz
	170 Hz	170 Hz	170 Hz	170 Hz	170 Hz	170 Hz
	Pan 1 CB-AD	Pan 2 CB-AD	Pan 3 CB-AD	Pan 4 CB-AD	Pan 5 CB-AD	Pan 6 CB-AD
5	99	99	99	99	99	99
	00	00	00	00	00	00
	Feedback 1	Feedback 2			Diffusion	
6	99%	99%	Inactive	Inactive	99	Inactive
	00	00			00	
	1	2	3	4	5	6
	Sliders					

* The software for this page allows fine tuning.

4.5

Splits—Bank 5

The five Split programs (Hall/Hall, Plate/Plate, Plate/Hall, Plate/Chorus, and Rich Split) allow the 224X to become two independent reverb units, with each unit processing a single input. The 224X processes two independent

reverb programs, each with its own variable parameters. To that end, the program's control pages are configured to accommodate individual inputs.

Table 4.19. Hall/Hall—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay L	Mid Decay L	Crossover L	Treble Decay L	Depth L > A	Predelay L
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	176 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	24.0 msec
2	LF Decay R	Mid Decay R	Crossover R	Treble Decay R	Depth R > B	Predelay R
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	176 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	24.0 msec
3	LF Stop Decay L	Mid Stop Decay L	Chorus LR	HF Bandwidth LR	Diffusion LR	Definition LR
	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
4	Preecho Level 1 L > A	Preecho Level 2 L > C	Preecho Level 3 R > B	Preecho Level 4 R > D	Inactive	Inactive
	99	99	99	99		
	00	00	00	00		
5*	Preecho Delay 1 L > A	Preecho Delay 2 L > C	Preecho Delay 3 R > B	Preecho Delay 4 R > D	Fine Predelay L > A	Fine Predelay R > B
	143 msec	143 msec	143 msec	143 msec	31.3 msec	31.3 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	1	2	3	4	5	6
Sliders						

*The software for this page allows fine tuning.

4.5.1 Hall/Hall—Program 1

This program processes each input with a Hall program similar to the Concert Hall program in bank 1. Table 4.19 lists the pages and variable parameter ranges. Note that the sliders on page 1 affect the signal entering the Left input and those on page 2 affect the Right input. In addition, the Dynamic Decay

toggle affects only the Left input. There are also four Preechoes (two for each channel).

This program has one variation, which sounds basically the same as variation 1 of the Concert Hall program.

Table 4.20. Plate/Plate—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay L	Mid Decay L	Crossover L	Treble Decay L	Attack L	Predelay L
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	176 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
2	LF Decay R	Mid Decay R	Crossover R	Treble Decay R	Attack R	Predelay R
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	176 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
3	LF Stop Decay L	Mid Stop Decay L	Chorus LR	HF Bandwidth LR	Diffusion LR	Definition LR
	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
4	Preecho Level 1 L > A	Preecho Level 2 L > C	Preecho Level 3 R > B	Preecho Level 4 R > D		
	99	99	99	99	Inactive	Inactive
	00	00	00	00		
5*	Preecho Delay 1 L > A	Preecho Delay 2 L > C	Preecho Delay 3 R > B	Preecho Delay 4 R > D	Fine Predelay L > A	Fine Predelay R > B
	170 msec	170 msec	170 msec	170 msec	30.2 msec	30.2 msec
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	1	2	3	4	5	6
Sliders						

*The software for this page allows fine tuning.

4.5.2 Plate/Plate—Program 2

This program processes each input with a Plate program similar to the Plate program in bank 3. Table 4.20 lists the control pages and variable parameter ranges. As in the Hall/Hall program, the first two pages of sliders control separate inputs, and the four Preechoes are also configured the same.

This program has two variations, with variation 1 basically the same as variation 1 of the Plate program, and variation 2 being a modified version with high predelay (left = 85.0 milliseconds and right = 69.0 milliseconds). Variation 2 also includes four preset Preecho Delay times and four preset Preecho Levels.

4.5.3 Plate/Hall—Program 3

This program processes the Left input with the Plate program and the Right input with the Concert Hall program. Table 4.21 lists the control pages and variable parameter ranges. The control pages are configured the same as the Hall/Hall and Plate/Plate programs.

This program has one variation, with the Left input being processed with a sound similar to variation 1 of the Plate program and the Right input being processed with a sound similar to variation 1 of the Concert Hall program.

4.5.4 Plate/Chorus—Program 4

This program combines two of the most generally useful programs in the 224X. With it you can combine chorusing and reverb on a single track or use the two sounds on different tracks at the same time. Table 4.22 lists the control pages and variable parameter ranges. The Left input is processed with the Plate program, and the Right input is processed with a modified Chorus & Echo program. There is only one variation.

As in the preceding Split programs, the Plate program retains virtually all of the features of the main Plate program, except that mode enhancement is inactive and therefore the Chorus slider has no effect (the Mode Enhancement toggle and the Chorus slider are active for the Chorus section of this program, however). In addition, this Plate program does not have Preechoes. The size of the plate is adjustable.

The Chorus program offers five voices (similar to the six voices in the Chorus & Echo program). Note that the HF Bandwidth and Diffusion controls apply to the Plate program only. Each voice in this Chorus program has full bandwidth and is not spread out by diffusion. Voices 1, 2, and 3 are in phase with the input and voices 4 and 5 are out of phase. The out-of-phase voices can be used to create a wonderful inverted flanging effect by setting their delays and pans within a few milliseconds of one or more of the noninverted voices. If cancellation is not desired, voices 4 and 5 can simply be spaced more than 7 milliseconds away from voices 1, 2, and 3.

A very interesting effect can be created with the Plate/Chorus program by panning all the voices in the Chorus program to one channel and using this output as the Left input feed to the Plate program—the resulting combination is excellent on musical material with long sustains, particularly synthesizer material.

Table 4.21. Plate/Hall—Control
Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
1	LF Decay L	Mid Decay L	Crossover L	Treble Decay L	Attack L	Predelay L
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	184 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
2	LF Decay R	Mid Decay R	Crossover R	Treble Decay R	Depth R>B	Predelay R
	70 sec	70 sec	19.0 kHz	19.0 kHz	99	184 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
3	LF Stop Decay L	Mid Stop Decay L	Chorus LR	HF Bandwidth LR	Diffusion LR	Definition LR
	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
4	Preecho Level 1 L>A	Preecho Level 2 L>C	Preecho Level 3 R>B	Preecho Level 4 R>D		
	99	99	99	99	Inactive	Inactive
	00	00	00	00		
5*	Preecho Delay 1 L>A	Preecho Delay 2 L>C	Preecho Delay 3 R>B	Preecho Delay 4 R>D		
	179 msec	179 msec	179 msec	179 msec	Inactive	Inactive
	0.00 msec	0.00 msec	0.00 msec	0.00 msec		
	1	2	3	4	5	6

*The software for this page allows fine tuning.

Table 4.22. Plate/Chorus—Control
Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	LF Decay L	Mid Decay L	Crossover L	Treble Decay L	Attack L	Predelay L
1	70 sec	70 sec	19.0 kHz	19.0 kHz	99	136 msec
	0.6 sec	0.6 sec	170 Hz	170 Hz	00	0.00 msec
	LF Stop Decay L	Mid Stop Decay L	Chorus R	HF Bandwidth L	Diffusion L	Definition L
2	70 sec	70 sec	97	19.0 kHz	99	75
	0.6 sec	0.6 sec	00	170 Hz	00	00
	Voice Level 1 R>	Voice Level 2 R>	Voice Level 3 R>	Voice Level 4 R>	Voice Level 5 R>	
3	99	99	99	99	99	Inactive
	00	00	00	00	00	
	Voice Delay 1 R>	Voice Delay 2 R>	Voice Delay 3 R>	Voice Delay 4 R>R>	Voice Delay 5	

Table 4.22. Plate/Chorus—Control
Pages and Variable Parameters cont'd.

Page	Variable Parameters and Ranges					
4*	430 msec	430 msec	430 msec	430 msec	430 msec	Inactive
	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	
	Feedback Gain 1 R>	Feedback Gain 2 R>	Feedback Gain 3 R>	Feedback Gain 4 R>	Feedback Gain 5 R>	
5	97%	97%	97%	97%	97%	Inactive
	–97%	–97%	–97%	–97%	–97%	
	Pan 1 B-D	Pan 2 B-D	Pan 3 B-D	Pan 4 B-D	Pan 5 B-D	
6	99	99	99	99	99	Inactive
	00	00	00	00	00	
	Size		Gate			
7	80 meters		5.08 sec			
	08 meters		0.00 sec			
	1	2	3	4	5	6
	Sliders					

*The software for this page allows fine tuning.

Table 4.23. Rich Split—Control Pages and Variable Parameters.

Page	Variable Parameters and Ranges					
	Left LF Decay	Left Mid Decay	Left Crossover	Left Treble Decay	Left Attack	Left Predelay
1	67 sec*	67 sec*	19.0 kHz*	19.0 kHz*	99	350 ms
	0.1 sec	0.1 sec	170 Hz	170 Hz	00	0.00 ms
2	Right LF Decay 67 sec*	Right Mid Decay 67 sec*	Right Crossover 19.0 kHz*	Right Treble Decay 19.0 kHz*	Right Attack 99	Right Predelay 350 ms
	0.1 sec	0.1 sec	170 Hz	170 Hz	00	000 ms
3	L LF Stop Decay	L Mid Stop Decay	L/R Chorus	L/R HF Bandwidth	L/R Diffusion	L/R Definition
	67 sec*	67 sec*	99	19 kHz*	99	99
	0.1 sec	0.1 sec	00	170 Hz	00	00
4	Preecho Level 1 L>A	Preecho Level 2 L>C	Preecho Level 3 R>B	Preecho Level 4 R>D	Inactive	Inactive
	99	99	99	99		
	00	00	00	00		
5	Preecho Delay 1 L>A	Preecho Delay 2 L>C	Preecho Delay 3 R>B	Preecho Delay 4 R>D	Fine Preecho Delay L>A	Fine Preecho Delay R>B
	125 ms	125 ms	125 ms	125 ms	125 ms	125 ms
	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms	0.00 ms
6	Left Size	Right Size	Reverb Stop Delay (Gate)	Inactive	Inactive	Inactive
	70 meters	70 meters	1.26 sec			
	08 meters	08 meters	0.00 sec			
	1	2	3	4	5	6
	Sliders					

*Can also be set to infinite.

4.5.5 Rich Split—Program 5

The Rich Split program provides two independent programs which sound similar to the Rich Chamber program. Each side of the Rich Split has a Size control, allowing the creation of two small rooms, a small room and a large room, two large rooms, or anything in between.

The Rich Split program has six control pages and only one variation. Table 4.23 lists the control pages and variable parameter ranges.

4.6

Program Block Diagrams

Fig. 4.1. Reverb: Concert Hall, Bright Hall, and Room.

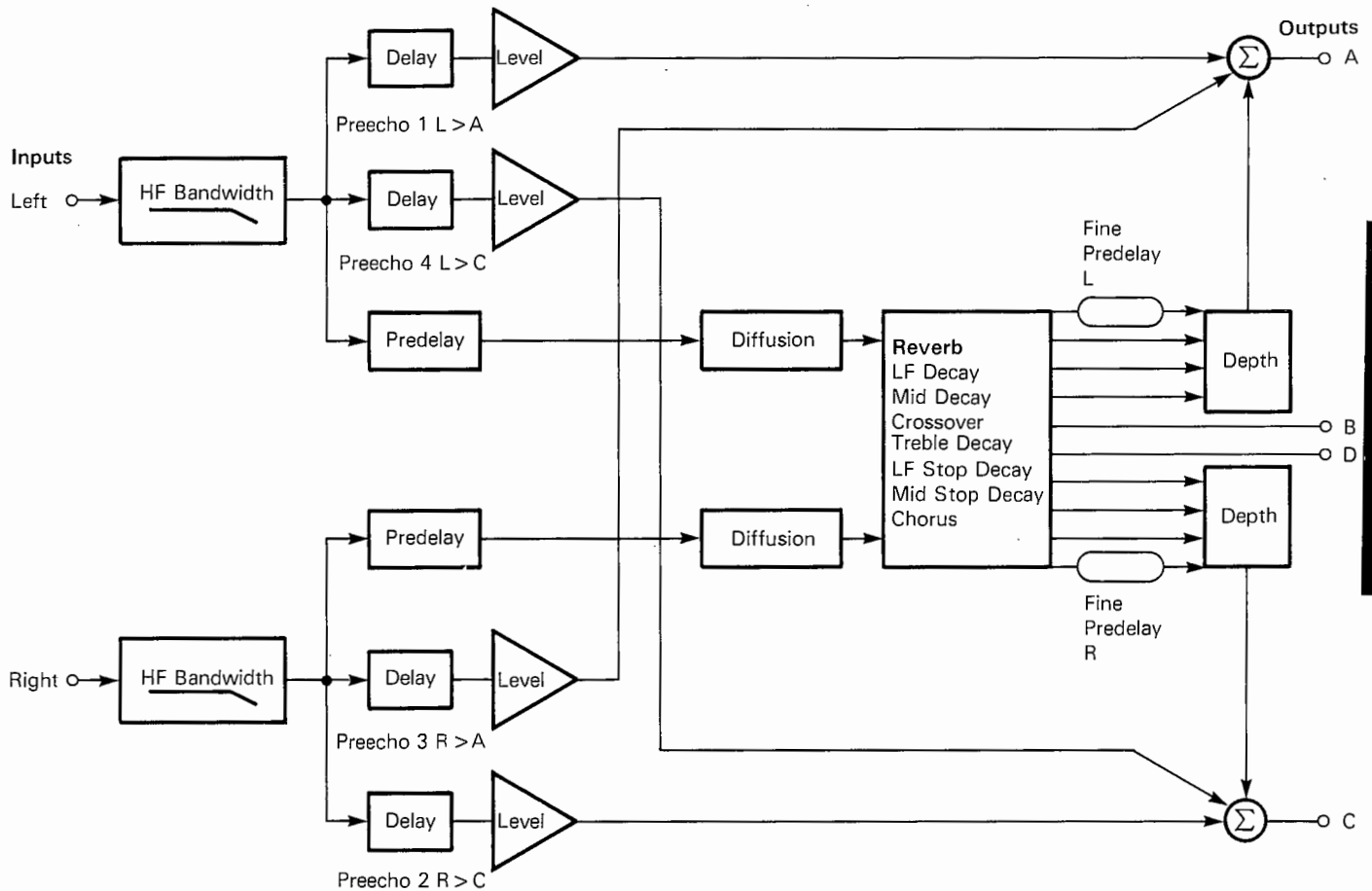


Fig. 4.2. Reverb: Dark Hall and Small Room.

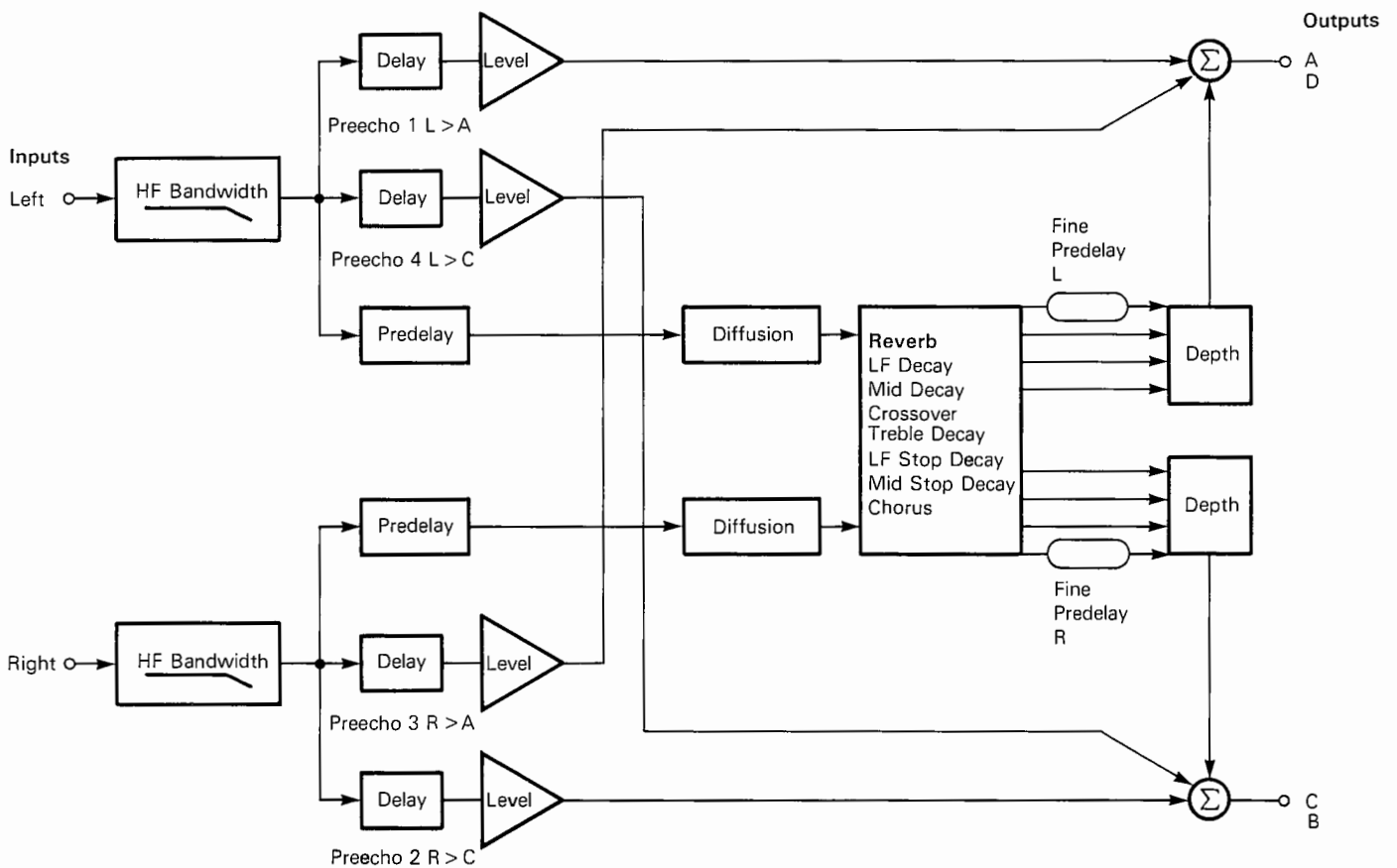


Fig. 4.3. Reverb: Chamber.

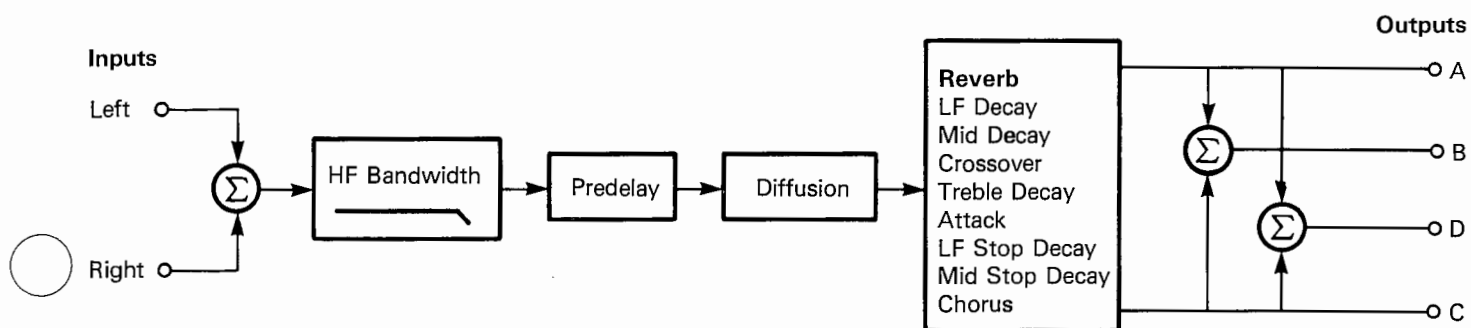


Fig. 4.4. Reverb: Rich Chamber,
Dark Chamber.

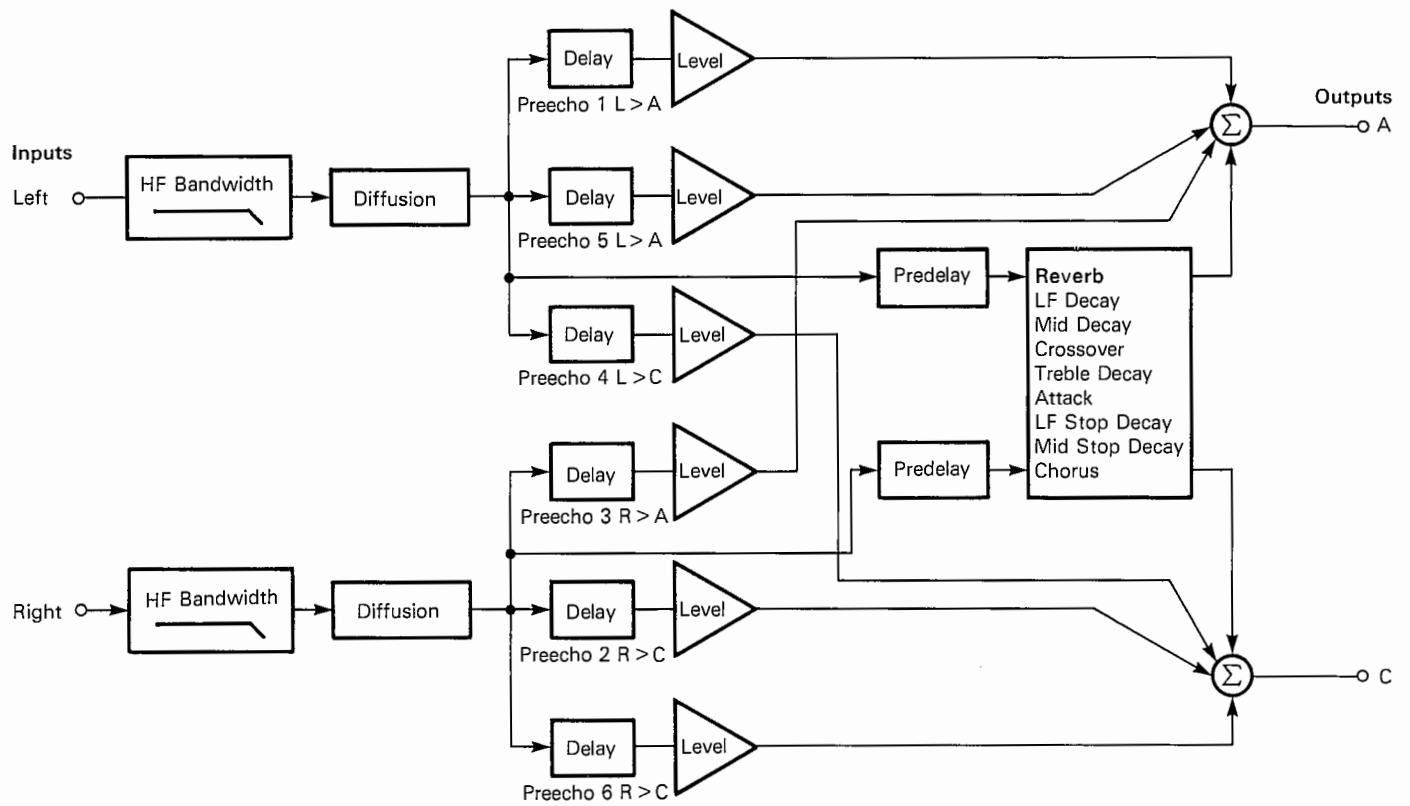


Fig. 4.5. Reverb: Plate, Small Plate, CD Plate A, CD Plate B, Rich Plate.

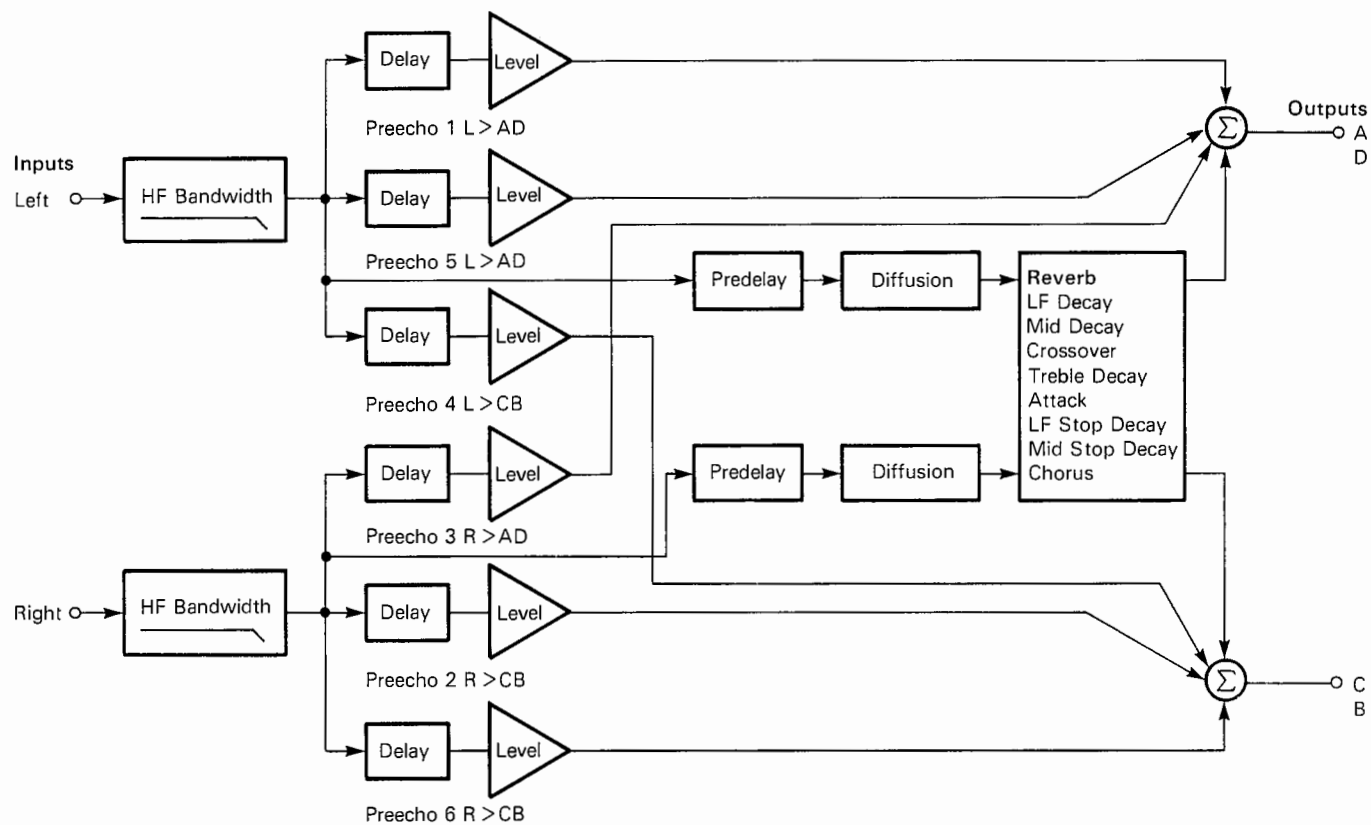


Fig. 4.6. Effects: Chorus & Echo.

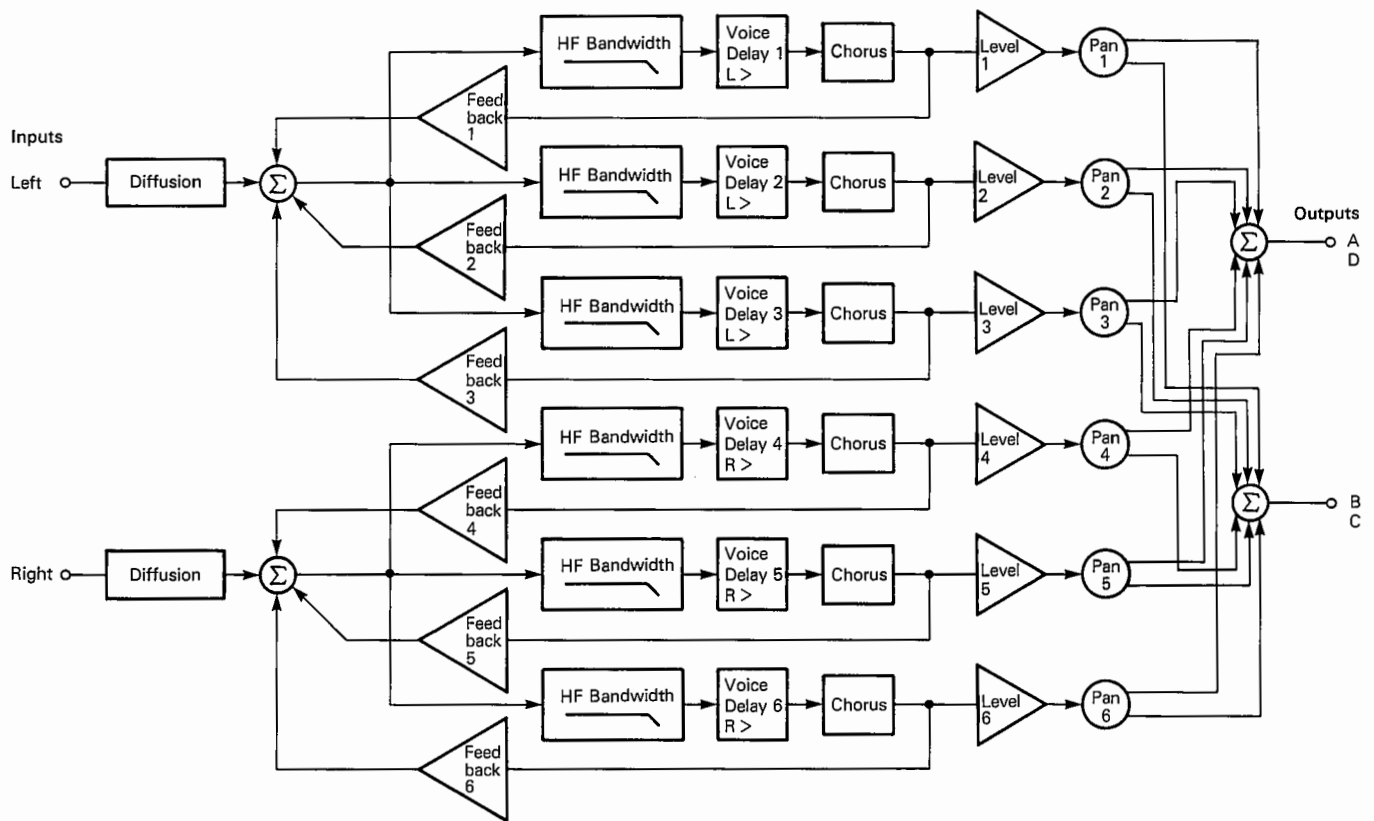


Fig. 4.7. Effects: Resonant Chords.

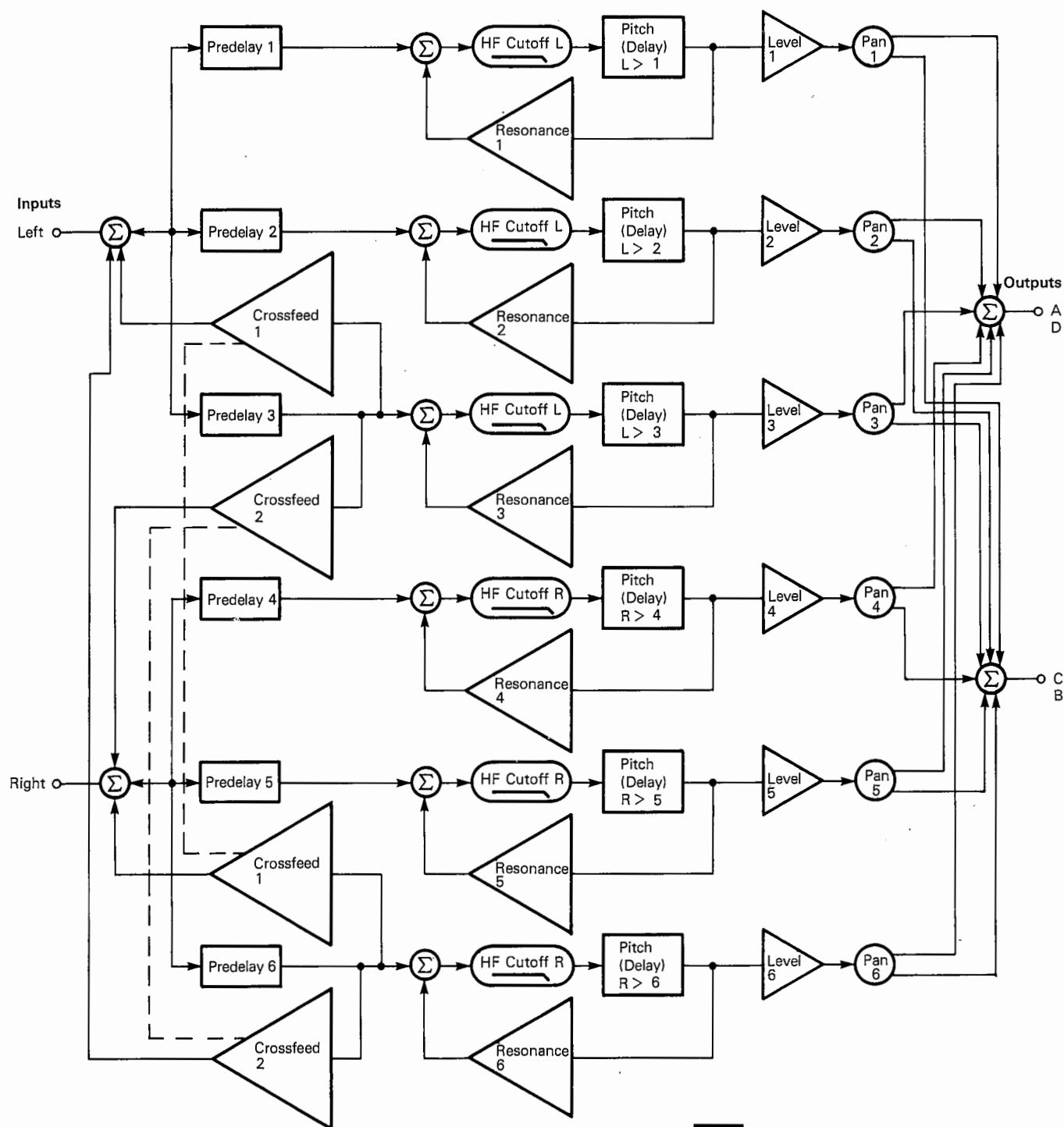


Fig. 4.8. Effects: Multiband Delay.

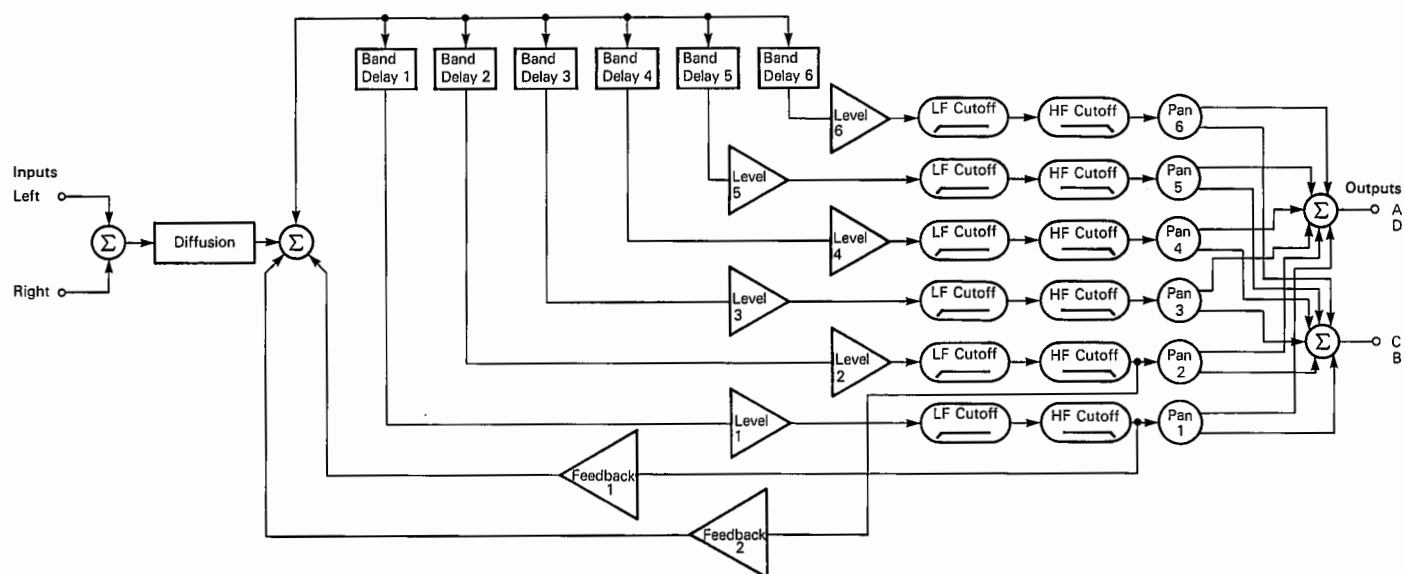


Fig. 4.9. Splits: Hall/Hall.

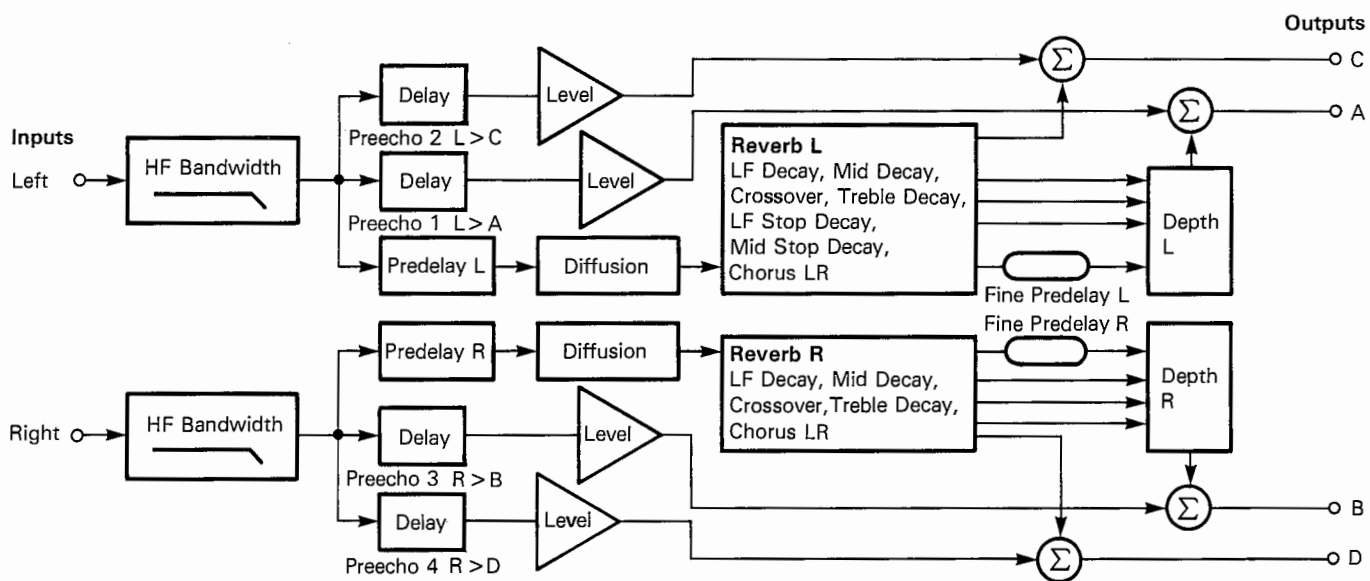


Fig. 4.10. Splits: Plate/Plate.

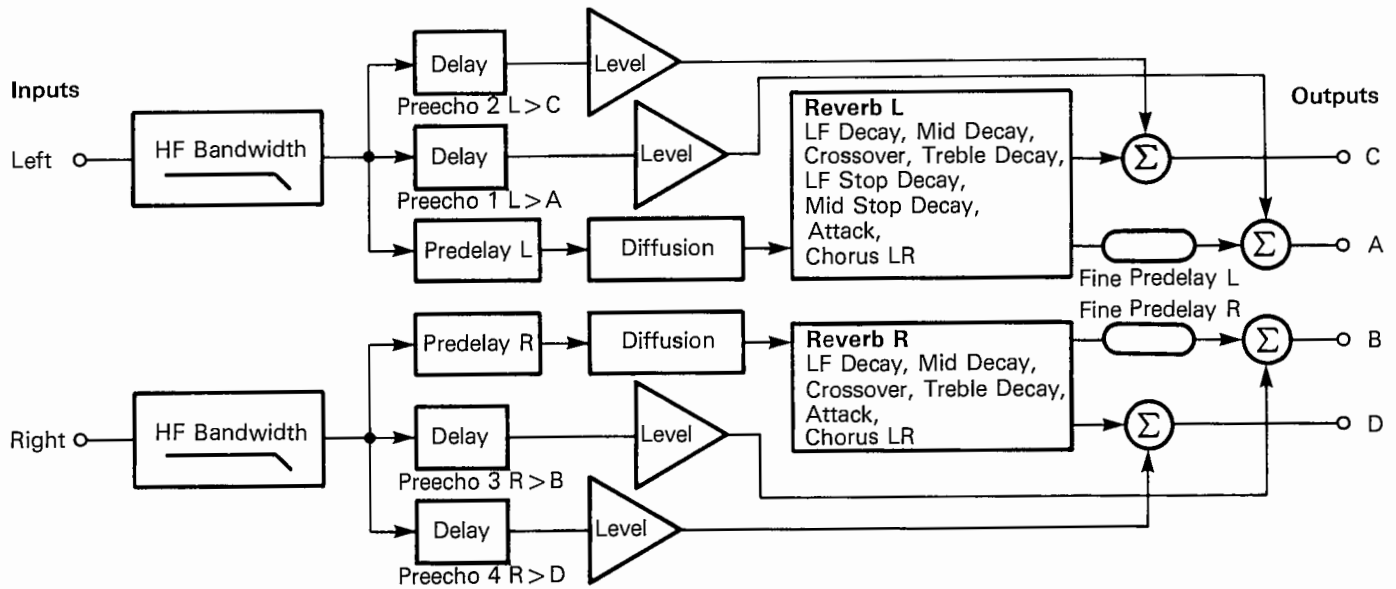


Fig. 4.11. Splits: Plate/Hall, Rich Split.

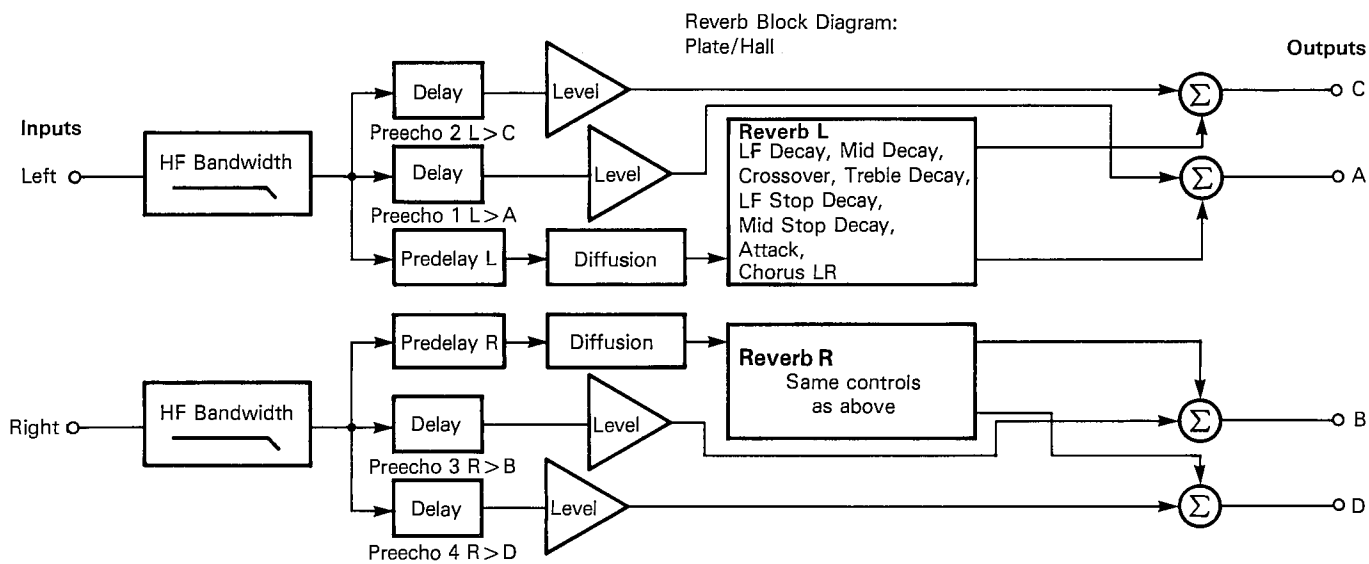
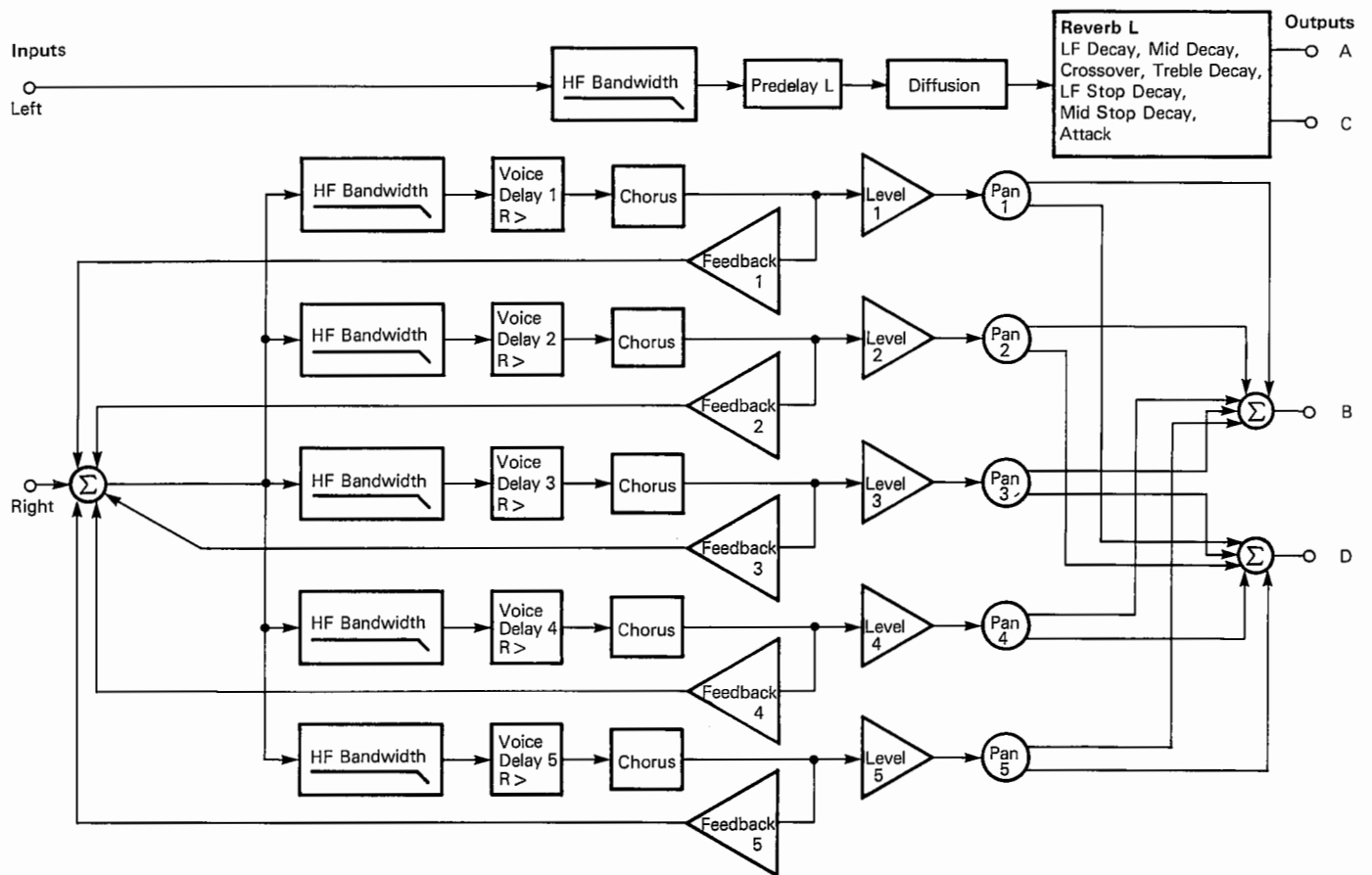


Fig. 4.12. Splits: Plate/Chorus.



This section gives a brief introduction to four general reverberation effects possible from the 224X and then gives some specific examples of how to achieve these effects. In addition, it describes applications of the 224X's special

effects programs and gives representative set-ups for those applications. Sonic shading and effects are highly subjective; these applications are meant to serve as general introductions and as starting points for experimentation.

5.1

Reverberation Programs

5.1.1 Re-creating Room Acoustics

The reverberation programs in the 224X can be used to add a realistic impression of a particular place to sonic material. This capability is obviously important for dialog and effects in broadcast or film, but realistic acoustics can be just as important in creating a mood to enhance a drum track or instrument. The choice of program depends on the mood wanted. Try the Rich Chamber or Dark Chamber first, adjusting the size and reverb time to match the material. Large sizes with predelay and preechoes sound very spacious,

and smaller sizes sound tight. The Small Room program is excellent for a mood of confined pressure, the Room program for a natural open sound, and the Concert Hall program for space or grandeur. When a hall is wanted for drums or popular music, try the Bright Hall program. For dialog or classical music, try the Dark Hall program. The Plate program, with predelays of about 100 milliseconds and preechoes (try variation 4), can sometimes be useful in emulating a large space.

5.1.2 Creating Space or Ambience

Ambience gives sonic material a sense of being real — of being actually performed in a particular place at a particular time. Very little reverberation need be used to create this illusion; no mudding or blurring of the material need take place. This use of the 224X is very similar to re-creating room acoustics, but larger spaces are usually used and not many

returns are used in the mix, so the particular space chosen is not obvious. This reverberation effect is commonly used for material that has already been mixed, such as classical music or a completed popular production. Again, the Bright Hall program is a good choice for drums or pop, and the Concert Hall or Dark Hall programs for classical music.

5.1.3 Thickening or Enriching Single Tracks

The bulk of reverberation in popular music production is used to thicken or enrich single tracks. Generally, the 224X Plate programs are best, because no definite sense of size or acoustic quality is wanted in a mixdown. Some engineers prefer the natural acoustic

quality of the Room program in a mix, and the Small Room or Rich Chamber programs can also produce interesting results. There are no rules for thickening with the 224X — even the Effects or Split programs can be useful.

5.1.4 Adding Loudness Without Increasing Peak Level

Closely miked material can often sound much softer than its peak recorded level would indicate, and an important use of reverberation is to increase loudness before cutting or broadcast. In this application, the Small Room program is very effective on dialog and on music when short reverb times are used.

A depth of about 60 gives more apparent level with less peak level. The Rich Chamber program is good when average reverb decay times of more than 1 second are used. Variation 3 of the Inverse Room program may also be very useful in adding loudness.

5.1.5 Examples of General Applications Using the Reverberation Programs with Specific Sources

Drums

1

Room Acoustics — try the Rich Chamber (variation 4), the Room (variation 1 or 2) and the Small Room (variations 1 and 2) programs. Also, the Bright Hall program has a bright sound and high enough diffusion to be successful with drums, and it gives a large, spacious sound.

2

Ambience — try the Rich Chamber or Rich Plate, with some pre-delay, or the Bright Hall program (variation 1 or 2); you don't need to add very much.

3

Thickening — try the Rich Plate program (variation 1 or 4). Set the low-frequency reverb decay time to less than 1 second, which can give a lighter sound. The Small Room program, with average reverb decay times of 0.6 second or less, simulates a drum cage, giving a much richer sound to a drum kit. The Constant-Density Plate B program produces a sound many engineers like on drums.

4

Adding Loudness — try the Rich Chamber, the Small Room program or the Inverse Room.

Vocals

1

Acoustics — try the Rich Chamber at different sizes, or the Concert Hall, Room, and Small Room programs (variation 1 or 2). The different sizes in these programs give quite different moods. Set the stopped reverb decay times (page 2, sliders 1 and 2) higher than running reverb decay times (page 1, sliders 1 and 2), and turn on the Dynamic Decay toggle (dynamic decay is preset on in variation 6 of the Concert Hall program). Variation 2 of the Concert Hall, Dark Hall, Bright Hall, Room, and Small Room programs has preset preechoes added to emulate the floor and back walls of a stage area. These variations increase the sense of distance between the performer and the pick-up.

2

Ambience — try the Rich Chamber, Concert Hall, or Dark Hall programs. Again, the preechoes in variation 2 may help.

3

Thickening — try the Rich Plate, Plate or Room programs. Variation 2 of the Plate program has low diffusion and a very interesting

clear sound on vocals. The Constant-Density Plate B program is also excellent. The Plate/Chorus program can be useful.

4

Loudness — try the Rich Chamber or Inverse Room program.

Dialog

1

Acoustics — dialog is very similar to vocals, but much less return should be added to the mix. The Small Room, Room, Dark Hall, and Rich Chamber programs can all be used, depending on the size and mood wanted. Variation 2 of these programs adds preechoes and gives a harder sound. Dynamic decay was originally developed for dialog and can be tried by turning on the Dynamic Decay Toggle.

2

Ambience — use the Dark Chamber, Dark Hall or Rich Chamber programs.

3

Thickening — try the Rich Chamber (small size), Small Room, Plate, Small Plate, Rich Chamber, or Constant-Density Plate B programs.

4

Loudness — try the Small Room, Rich Chamber, or Inverse Room programs.

Orchestra

1

Acoustics — a small group might sound more natural using the Room and Small Room programs, but the Dark Chamber program should almost always be tried first. Variation 2 adds preechoes that sound natural on symphonic or vocal material. The Rich Chamber program is also valuable for this application; the predelay can be set to 120 milliseconds, and a few preechoes can be used to fill in the time between the input and the reverb. This configuration gives a good sense of space with less color than is available from the Dark Hall program.

2

Ambience — the Dark Hall program gives the most realistic sense of space.

3

Thickening — try the Rich Chamber program with short predelay and with average reverb decay times of 1 to 2 seconds.

4

Loudness — try the Rich Chamber program.

The 224X's Effects programs are more variable and even more subjective than the reverberation programs. Applications depend in large part on musical styles, context, and the intent of an artist or producer.

Although the 224X Effects programs can produce an enormous array of different sounds, many of the common uses can be grouped into a few general classes of functions. These functions consist of:

- Enrichment or Thickening
- Doubling, Chorusing, and Flanging
- Repeats
- Sound Modification

The following sections describe a few applications for these functions.

5.2.1 Enrichment or Thickening

In contemporary production, much attention is given to achieving the quality of fullness in sound. Each of the effects programs can increase the fullness or richness of musical parts. The extraordinary amount of variability within each program allows a broad palette of distinctive sounds. For instance, it is very easy to take a simple chorus and add repeats by increasing the delay time of one of the delay taps, or "voices." In addition, the independent processing of inputs (Chorus & Echo, Resonant Chords, and Plate/Chorus) allows a user to create more than one effect simultaneously.

The general form of an "enrichment" setup is one or more short delay taps combined with the original signal. The delay times selected depend on the type of sound desired and the input material. Very short delays are likely to affect the tone quality of the source through phase cancellation. If the delays are too long, they are heard as discrete repeats or "slaps." Percussive material tends to accentuate the discreteness of the delays, but slaps can be smoothed with the Diffusion control in the Chorus & Echo and Multiband Delay programs. Vocals or other nonpercussive inputs can use longer delay times and lower diffusion.

The number of delay taps that is mixed in (level settings above 0) also has a marked effect on the type of sound achieved. Generally speaking, the greater the number of delay taps brought into play, the thicker the sound. Phase cancellations between the taps can be used to color the sound, and panning the various taps to different points in the stereo field can create a dramatic sense of dimensionality.

Recirculation of the delays forms a resonant "comb" filter and colors the sound (an effect exploited by the Resonant Chords program). The result can be both interesting and bizarre.

The Chorus & Echo, Resonant Chords, and Plate/Chorus programs can also be used in "series" by routing the outputs of one side into the input of the other. The results can be very interesting — for example, the Plate/Chorus program can add reverberation to all voices of a 5-voice chorus, giving a combination that is particularly effective on synthesizer or vocals.

The Chorus & Echo program is most effective when an element of motion is desired, as in actual chorusing or flanging (see Sec. 5.2.2). The Resonant Chords program can be used for enrichment if the Resonance parameter for each tap is reduced. The Multiband Delay program can be used to delay different frequency bands by different amounts, creating richness without comb-filter effects. The Plate/Chorus program is useful when an element of room acoustics is desired along with or on top of the thickening function.

Tables 5.1 to 5.4 show possible setups for thickening, using each of the Effects programs. These setups represent just four of thousands of possible, and useful, enrichment setups. Use the values detailed in the tables as starting points to create your own distinct sounds. With the memory storage and cassette interface of the 224X with LARC, an engineer, artist, or producer can create a library of setups that can be used in the future.

Table 5.1. Enrichment 1 – Chorus & Echo Program.

Page	Variable Parameters and Ranges					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	69	19.0 kHz	00	Inactive
	Voice Level 1 L>	Voice Level 2 L>	Voice Level 3 L>	Voice Level 4 R>	Voice Level 5 R>	Voice Level 6 R>
2	56	56	50	61	41	48
	Voice Delay 1 L>	Voice Delay 2 L>	Voice Delay 3 L>	Voice Delay 4 R>	Voice Delay 5 R>	Voice Delay 6 R>
3	12.0 msec	23.0 msec	38.0 msec	7.00 msec	10.0 msec	33.0 msec
	Feedback 1 L>	Feedback 2 L>	Feedback 3 L>	Feedback 4 R>	Feedback 5 R>	Feedback 6 R>
4	00%	00%	00%	– 28%	– 12%	09%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	50	37	00	99
	1	2	3	4	5	6
	Sliders					

Table 5.2. Enrichment 2 — Resonant
Chords Program.

Page	Variable Parameters and Ranges					
	Note Level 1 L>	Note Level 2 L>	Note Level 3 L>	Note Level 4 R>	Note Level 5 R>	Note Level 6 R>
1	69	45	41	58	45	56
	Note Pitch 1 L>	Note Pitch 2 L>	Note Pitch 3 L>	Note Pitch 4 R>	Note Pitch 5 R>	Note Pitch 6 R>
2	4.40 msec	5.87 msec	6.96 msec	8.75 msec	11.6 msec	17.3 msec
	Resonance 1 L>	Resonance 2 L>	Resonance 3 L>	Resonance 4 R>	Resonance 5 R>	Resonance 6 R>
3	50%	53%	28%	28%	47%	59%
	Predelay 1 L>	Predelay 2 L>	Predelay 3 L>	Predelay 4 R>	Predelay 5 R>	Predelay 6 R>
4	12.0 msec	24.0 msec	36.0 msec	28.0 msec	56.0 msec	44.0 msec
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	31	89	06	58
	Crossfeed	Crossfeed	HF Cutoff L	HF Cutoff R		
6	00%	00%	4.90 kHz	11.3 kHz	Inactive	Inactive
	1	2	3	4	5	6
	Sliders					

Table 5.3. Enrichment 3 — Multiband Delay Program.

Page	Variable Parameter Settings					
	Band Level 1 L + R >	Band Level 2 L + R >	Band Level 3 L + R >	Band Level 4 L + R >	Band Level 5 L + R >	Band Level 6 L + R >
1	50	50	50	50	50	50
	Band Delay 1 L + R >	Band Delay 2 L + R >	Band Delay 3 L + R >	Band Delay 4 L + R >	Band Delay 5 L + R >	Band Delay 6 L + R >
2	6.00 msec	16.0 msec	12.0 msec	16.0 msec	27.0 msec	36.0 msec
	LF Cutoff 1 L + R >	LF Cutoff 2 L + R >	LF Cutoff 3 L + R >	LF Cutoff 4 L + R >	LF Cutoff 5 L + R >	LF Cutoff 6 L + R >
3	170 Hz	350 Hz	530 Hz	1.13 kHz	3.80 kHz	19.0 kHz
	HF Cutoff 1 L + R >	HF Cutoff 2 L + R >	HF Cutoff 3 L + R >	HF Cutoff 4 L + R >	HF Cutoff 5 L + R >	HF Cutoff 6 L + R >
4	170 Hz	350 Hz	530 Hz	1.13 kHz	3.80 kHz	19.0 kHz
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	06	28	52	74	91
	Feedback 1	Feedback 2			Diffusion	
6	00	00	Inactive	Inactive	33	Inactive
	1	2	3	4	5	6
	Sliders					

Table 5.4. Enrichment 4 — Plate/
Chorus Program.

Page	Variable Parameter Settings					
	LF Decay L	Mid Decay L	Crossover L	Treble Decay L	Attack L	Predelay L
1	0.6 sec	0.6 sec	1.13 kHz	19.0 kHz	00	0.00 msec
	LF Stop Decay L	Mid Stop Decay L	Chorus R	HF Bandwidth L	Diffusion L	Definition L
2	3.0 sec	3.0 sec	44	9.0 kHz	28	00
	Voice Level 1	Voice Level 2	Voice Level 3	Voice Level 4	Voice Level 5	
	R >	R >	R >	R >	R >	
3	50	50	44	44	37	Inactive
	Voice Delay 1 R >	Voice Delay 2 R >	Voice Delay 3 R >	Voice Delay 4 R >	Voice Delay 5 R >	
4	5.50 msec	7.00 msec	3.50 msec	5.50 msec	9.00 msec	Inactive
	Feedback 1 R >	Feedback 2 R >	Feedback 3 R >	Feedback 4 R >	Feedback 5 R >	
5	09%	— 06%	06%	— 03%	00%	Inactive
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	
6	60	99	25	75	50	Inactive
	1	2	3	4	5	6
	Sliders					

5.2.2 Doubling, Chorus, and Flanging

Double tracking and chorusing are perhaps the most popular of all effects. Although they might be classed as forms of enrichment, they are distinct enough to merit their own discussion. In double tracking (also known as doubling or artificial double tracking), the intent is to create a realistic imitation of the effect of a sound actually being "doubled" on tape.

The Chorus & Echo program provides two separate channels for doubling. The basic doubling setup consists of one delay tap set for between 15 and 30 milliseconds delay, with some modulation of the delay time. This delay tap is usually panned to the opposite side of the stereo field from the original. The Chorus parameter should be set moderately high, and no feedback should be used. Table 5.5 depicts a doubling setup for one side of the Chorus & Echo program. The other side (R input and voices 4 to 6) can be set up for an independent double, or for an entirely different effect.

Table 5.5. Doubling — Chorus & Echo Program.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	97	19.0 kHz	00	Inactive
	Voice Level 1 L>	Voice Level 2 L>	Voice Level 3 L>	Voice Level 4 R>	Voice Level 5 R>	Voice Level 6 R>
2	80	00	00	00	00	00
	Voice Delay 1 L>	Voice Delay 2 L>	Voice Delay 3 L>	Voice Delay 4 R>	Voice Delay 5 R>	Voice Delay 6 R>
3	18.0 msec					
	Feedback 1 L>	Feedback 2 L>	Feedback 3 L>	Feedback 4 R>	Feedback 5 R>	Feedback 6 R>
4	00%					
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00					
	1	2	3	4	5	6
	Sliders					

Chorus is similar to doubling but uses a number of voices. It is very popular to exaggerate the chorus effect, so that it becomes more swirling and artificial. To achieve this effect, bring in lots of voices, turn up the modulation (Chorus parameter), place the voices very close together in time, and add recirculation around the delays. The effects can be dramatic. Table 5.6 shows a moderately thick chorus setup, somewhere between a realistic chorus and a totally artificial sound.

Table 5.6. Chorusing — Chorus & Echo Program.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	97	19.0 kHz	00	Inactive
	Voice Level 1 L >	Voice Level 2 L >	Voice Level 3 L >	Voice Level 4 R >	Voice Level 5 R >	Voice Level 6 R >
2	56	56	50	50	44	44
	Voice Delay 1 L >	Voice Delay 2 L >	Voice Delay 3 L >	Voice Delay 4 R >	Voice Delay 5 R >	Voice Delay 6 R >
3	18.0 msec	22.0 msec	17.0 msec	19.0 msec	17.0 msec	18.0 msec
	Feedback 1 L >	Feedback 2 L >	Feedback 3 L >	Feedback 4 R >	Feedback 5 R >	Feedback 6 R >
4	34%	00%	-25%	37%	00%	-31%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	1	2	3	4	5	6
	Sliders					

Flanging effects result from having very short delays mixed together while time modulation (Chorus parameter) is in effect. This results in shifting "comb filters" in which some frequencies are canceled and some are reinforced. The motion of these notches and peaks creates the familiar "jet plane" sound.

In the Chorus & Echo program, the six output voices or delay taps, each with an independent random modulation, provide the opportunity to achieve complex flanging motion. When the taps are set for identical values, the modulations create motion around that point, and adding recirculation exaggerates the effect. By setting the delays for larger values, flanging effects can be combined with other effects, such as chorus or repeats. Tables 5.7 and 5.8 show two possibilities. Table 5.7 illustrates a basic flange, and Table 5.8 shows a flange that combines with multiple repeats for a very pronounced, swirling echo.

Table 5.7. Basic Flange.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	97	19.0 kHz	18	Inactive
	Voice Level 1 L >	Voice Level 2 L >	Voice Level 3 L >	Voice Level 4 R >	Voice Level 5 R >	Voice Level 6 R >
2	50	50	50	50	50	50
	Voice Delay 1 L >	Voice Delay 2 L >	Voice Delay 3 L >	Voice Delay 4 R >	Voice Delay 5 R >	Delay 6 R >
3	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Feedback 1 L >	Feedback 2 L >	Feedback 3 L >	Feedback 4 R >	Feedback 5 R >	Feedback 6 R >
4	37%	- 25%	- 25%	37%	- 22%	- 31%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	1	2	3	4	5	6
Sliders						

Table 5.8. Flange with Multiple Repeats.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	97	19.0 kHz	44	Inactive
	Voice Level 1 L>	Voice Level 2 L>	Voice Level 3 L>	Voice Level 4 R>	Voice Level 5 R>	Voice Level 6 R>
2	50	50	50	50	50	50
	Voice Delay 1 L>	Voice Delay 2 L>	Voice Delay 3 L>	Voice Delay 4 R>	Voice Delay 5 R>	Voice Delay 6 R>
3	300 msec	300 msec	300 msec	300 msec	300 msec	300 msec
	Feedback 1 L>	Feedback 2 L>	Feedback 3 L>	Feedback 4 R>	Feedback 5 R>	Feedback 6 R>
4	44%	–31%	–25%	44%	–22%	–04%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	1	2	3	4	5	6
	Sliders					

5.2.3 Repeats

When delays exceed a few tens of milliseconds, discrete repetitions are heard. The multiple delay outputs of the Effects programs provide many opportunities for complex patterns of repeats. Each of the Effects programs can provide up to six distinct repeats, and recirculation, or feedback, can stretch out these repeats almost indefinitely. By using the distinct nature of each program, these repeats, or echoes, can be combined with various other types of effects with startling results. Tables 5.9 to 5.11 show just a few possible settings

Table 5.9. Repeats 1 — Chorus & Echo Program.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	53	19.0 kHz	25	Inactive
	Voice Level 1 L>	Voice Level 2 L>	Voice Level 3 L>	Voice Level 4 R>	Voice Level 5 R>	Voice Level 6 R>
2	99	00	00	99	00	00
	Voice Delay 1 L>	Voice Delay 2 L>	Voice Delay 3 L>	Voice Delay 4 R>	Voice Delay 5 R>	Voice Delay 6 R>
3	750 msec			798 msec		
	Feedback 1 L>	Feedback 2 L>	Feedback 3 L>	Feedback 4 R>	Feedback 5 R>	Feedback 6 R>
4	41%	00%	00%	41%	00%	00%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	1	2	3	4	5	6
	Sliders					

Table 5.10. Repeats 2 — Chorus & Echo Program.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	53	19.0 kHz	25	Inactive
	Voice Level 1 L>	Voice Level 2 L>	Voice Level 3 L>	Voice Level 4 R>	Voice Level 5 R>	Voice Level 6 R>
2	99	69	67	99	72	63
	Voice Delay 1 L>	Voice Delay 2 L>	Voice Delay 3 L>	Voice Delay 4 R>	Voice Delay 5 R>	Voice Delay 6 R>
3	750 msec	693 msec	728 msec	798 msec	644 msec	705 msec
	Feedback 1 L>	Feedback 2 L>	Feedback 3 L>	Feedback 4 R>	Feedback 5 R>	Feedback 6 R>
4	41%	28%	00%	41%	25%	00%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	1	2	3	4	5	6
	Sliders					

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Table 5.11. Repeats 3 — Resonant Chords Program.

Page	Variable Parameter Settings					
	Note Level 1 L >	Note Level 2 L >	Note Level 3 L >	Note Level 4 R >	Note Level 5 R >	Note Level 6 R >
1	50	50	50	50	50	50
	Note Pitch 1 L >	Note Pitch 2 L >	Note Pitch 3 L >	Note Pitch 4 R >	Note Pitch 5 R >	Note Pitch 6 R >
2	4.40 msec	5.87 msec	6.96 msec	8.75 msec	11.6 msec	17.3 msec
	Resonance 1 L >	Resonance 2 L >	Resonance 3 L >	Resonance 4 R >	Resonance 5 R >	Resonance 6 R >
3	75%	— 78%	59%	— 56%	85%	— 50%
	Predelay 1 L >	Predelay 2 L >	Predelay 3 L >	Predelay 4 R >	Predelay 5 R >	Predelay 6 R >
4	192 msec	368 msec	400 msec	864 msec	364 msec	504 msec
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	25	66	15	88	25	96
	Crossfeed	Crossfeed	HF Cutoff L	HF Cutoff R		
6	00%	61%	19.0 kHz	19.0 kHz	Inactive	Inactive
	1	2	3	4	5	6
	Sliders					

Table 5.12. Repeats 4 — Multiband Delay Program.

Page	Variable Parameter Settings					
	Band Level 1 L + R >	Band Level 2 L + R >	Band Level 3 L + R >	Band Level 4 L + R >	Band Level 5 L + R >	Band Level 6 L + R >
1	50	50	50	50	50	50
	Band Delay 1 L + R >	Band Delay 2 L + R >	Band Delay 3 L + R >	Band Delay 4 L + R >	Band Delay 5 L + R >	Band Delay 6 L + R >
2	568 msec	1.28 sec	1.16 sec	1.44 sec	1.24,sec	1.86 sec
	LF Cutoff 1 L + R >	LF Cutoff 2 L + R >	LF Cutoff 3 L + R >	LF Cutoff 4 L + R >	LF Cutoff 5 L + R >	LF Cutoff 6 L + R >
3	720 Hz	1.13 kHz	5.30 kHz	1.13 kHz	1.57 kHz	3.40 kHz
	HF Cutoff 1 L + R >	HF Cutoff 2 L + R >	HF Cutoff 3 L + R >	HF Cutoff 4 L + R >	HF Cutoff 5 L + R >	HF Cutoff 6 L + R >
4	1.80 kHz	920 Hz	530 Hz	1.13 kHz	6.90 kHz	4.90 kHz
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	Feedback 1	Feedback 2			Diffusion	
6	22%	20%	Inactive	Inactive	20	Inactive
	1	2	3	4	5	6
	Sliders					

5.2.4 Sound Modification

Sound modification is a very broad term, meaning that something is done to a sound to turn it into something new. All of the effects mentioned so far are actually sound modifications, but this section discusses more exaggerated forms that alter the basic nature of the input signal.

The Resonant Chords program provides a dramatic transformation. A percussive source is altered by using transients to "ring" a group of resonators, each consisting of a short delay being fed back strongly on itself. For each note, the pitch, duration, tone, and initial

delay of the sound can be adjusted; the various notes can be adjusted into different, related chords; and the Note Predelay parameter can be tuned to match a particular tempo. If these different tunings and adjustments are stored in memory registers, a drum track can be played into the unit and changed into a complete backing track.

The other programs can also be used to produce startling, unnatural effects for special purposes. Tables 5.13 to 5.16 show a few of the possibilities.

Table 5.13. Modification 1 — Chorus & Echo.

Page	Variable Parameter Settings					
			Chorus	HF Bandwidth	Diffusion	
1	Inactive	Inactive	97	10.1 kHz	00	Inactive
	Voice Level 1 L >	Voice Level 2 L >	Voice Level 3 L >	Voice Level 4 R >	Voice Level 5 R >	Voice Level 6 R >
2	56	56	50	50	44	44
	Voice Delay 1 L >	Voice Delay 2 L >	Voice Delay 3 L >	Voice Delay 4 R >	Voice Delay 5 R >	Voice Delay 6 R >
3	0.00 msec	178 msec	15.0 msec	0.00 msec	45.0 msec	19.0 msec
	Feedback 1 L >	Feedback 2 L >	Feedback 3 L >	Feedback 4 R >	Feedback 5 R >	Feedback 6 R >
4	97%	00%	00%	97%	00%	00%
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	1	2	3	4	5	6
Sliders						

Table 5.14. Modification 2 —
Resonant Chords Program.

Page	Variable Parameter Settings					
	Note Level 1 L>	Note Level 2 L>	Note Level 3 L>	Note Level 4 R>	Note Level 5 R>	Note Level 6 R>
1	50	50	50	50	50	50
	Note Pitch 1 L>	Note Pitch 2 L>	Note Pitch 3 L>	Note Pitch 4 R>	Note Pitch 5 R>	Note Pitch 6 R>
2	4.40 msec	5.87 msec	6.96 msec	8.75 msec	11.6 msec	17.3 msec
	Resonance 1 L>	Resonance 2 L>	Resonance 3 L>	Resonance 4 R>	Resonance 5 R>	Resonance 6 R>
3	97%	97%	97%	97%	97%	97%
	Predelay 1 L>	Predelay 2 L>	Predelay 3 L>	Predelay 4 R>	Predelay 5 R>	Predelay 6 R>
4	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	Crossfeed	Crossfeed	HF Cutoff L	HF Cutoff R		
6	00%	00%	4.10 kHz	4.10 kHz	Inactive	Inactive
	1	2	3	4	5	6
	Sliders					

Table 5.15. Modification 3 —
Resonant Chords Program.

Page	Variable Parameter Settings					
	Note Level 1 L >	Note Level 2 L >	Note Level 3 L >	Note Level 4 R >	Note Level 5 R >	Note Level 6 R >
1	50	50	50	50	50	50
	Note Pitch 1 L >	Note Pitch 2 L >	Note Pitch 3 L >	Note Pitch 4 R >	Note Pitch 5 R >	Note Pitch 6 R >
2	4.93 msec	6.59 msec	7.89 msec	9.84 msec	13.0 msec	19.8 msec
	Resonance 1 L >	Resonance 2 L >	Resonance 3 L >	Resonance 4 R >	Resonance 5 R >	Resonance 6 R >
3	97%	97%	97%	97%	97%	97%
	Predelay 1 L >	Predelay 2 L >	Predelay 3 L >	Predelay 4 R >	Predelay 5 R >	Predelay 6 R >
4	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	Crossfeed	Crossfeed	HF Cutoff L	HF Cutoff R		
6	00%	00%	4.10 kHz	4.10 kHz	Inactive	Inactive
	1	2	3	4	5	6
	Sliders					

Table 5.16. Modification 4 —
Resonant Chords Program.

Page	Variable Parameter Settings					
	Note Level 1 L >	Note Level 2 L >	Note Level 3 L >	Note Level 4 R >	Note Level 5 R >	Note Level 6 R >
1	50	50	50	50	50	50
	Note Pitch 1 L >	Note Pitch 2 L >	Note Pitch 3 L >	Note Pitch 4 R >	Note Pitch 5 R >	Note Pitch 6 R >
2	3.68 msec	4.93 msec	5.90 msec	7.34 msec	9.71 msec	14.7 msec
	Resonance 1 L >	Resonance 2 L >	Resonance 3 L >	Resonance 4 R >	Resonance 5 R >	Resonance 6 R >
3	97%	97%	97%	97%	97%	97%
	Predelay 1 L >	Predelay 2 L >	Predelay 3 L >	Predelay 4 R >	Predelay 5 R >	Predelay 6 R >
4	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec	0.00 msec
	Pan 1	Pan 2	Pan 3	Pan 4	Pan 5	Pan 6
5	00	99	00	99	00	99
	Crossfeed	Crossfeed	HF Cutoff L	HF Cutoff R		
6	00%	00%	4.10 kHz	4.10 kHz	Inactive	Inactive
	1	2	3	4	5	6
	Sliders					